UNIT - I

LAMINAR & TURBULENT FLOW IN PIPES

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

- Laminar flow through circular pipes, annulus and parallel plates
- Stokes' law; measurement of viscosity
- Reynolds experiment; transition from laminar to turbulent flow
- Resistance to flow in smooth and rough pipes; Moody's diagram
- Introduction to boundary layer theory

1. Basics: Viscosity and Types of Flow

1.1 Newton's law of viscosity

Shear stress between adjacent fluid layers is proportional to the **velocity gradient**.

$$\tau = \mu \, (du/dy)$$

- μ = dynamic viscosity (Pa.s or $N.s/m^2$; 1 poise = 0.1 Pa.s). Kinematic viscosity $\nu = \mu/\rho$ (m^2/s; 1 stoke = 10^{-4} m^2/s).
- Viscosity of **liquids decreases** with temperature; of **gases increases**.

1.2 Laminar and turbulent flow

Basis	Laminar flow	Turbulent flow
Path of particles	Smooth, parallel layers (streamlines)	Irregular, zig-zag, mixing, eddies
Shear stress	Only viscous: $\tau = \mu \, du/dy$	Viscous + turbulent (Reynolds) stress
Velocity profile	Parabolic	Flatter, fuller near the core
Reynolds number	Re < 2000 (pipes)	Re > 4000 (pipes)
Head loss	hf V	hf V^1.75 to 2
Occurs in	Viscous fluids, low velocity, small pipes	Water, air at ordinary velocities

1.3 Reynolds number

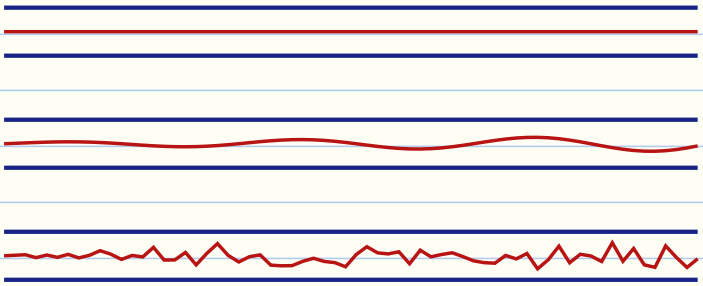
$$Re = \rho V D / \mu = V D / \nu$$

- Ratio of **inertia force** to **viscous force**. For pipes: laminar if **Re < 2000**, transition 2000 to 4000, turbulent if **Re > 4000**.

2. Reynolds Experiment

2.1 Set-up and procedure

- A large glass tank of water, a horizontal glass tube with a bell-mouthed entry and a regulating valve; a **dye container** injects a fine filament of coloured dye at the entry.



Laminar flow (Re < 2000)

Transition (2000 < Re < 4000)

Turbulent flow (Re > 4000)

- At **low velocity** the dye moves as a straight thread (laminar). As velocity is raised the thread starts to wave (transition), and at **high velocity** it breaks up and mixes completely (turbulent).

2.2 Observations

- **Lower critical velocity**: velocity (Re about 2000) below which flow is always laminar. **Upper critical velocity**: Re about 4000 (up to 10,000 in very quiet conditions) above which flow is turbulent.
- Transition: flow is unstable; small disturbances make it turbulent. Depends on **roughness, entry conditions and disturbances**.

2.3 Critical velocity

$$V_c = Re_c \cdot \nu / D \quad \text{(for pipe)}$$

Example: Water $\nu = 10^{-6}$ m²/s flows in D = 25 mm pipe. Lower critical velocity = 2000 x 10^{-6} / 0.025 = **0.08 m/s**.

3. Laminar Flow Through a Circular Pipe (Hagen-Poiseuille)

3.1 Assumptions

- Steady, incompressible, fully developed laminar flow; pipe horizontal and long; no slip at wall.

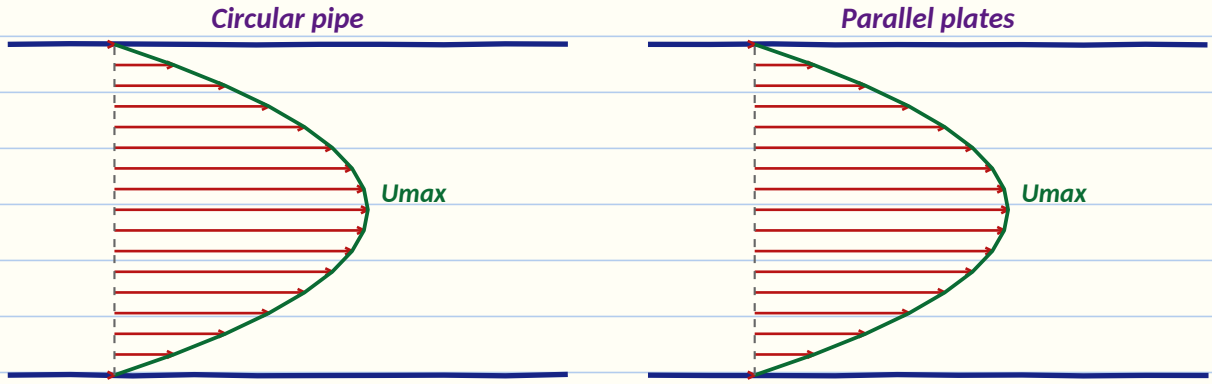
3.2 Derivation (summary)

- Take a coaxial fluid cylinder of radius r and length dx. **Pressure force = shear force** :

$$\tau = -(\partial p/\partial x)(r/2) \quad \text{(shear stress varies linearly: zero at axis, maximum at wall)}$$

- Using $\tau = -\mu\, du/dr$ and integrating with $u = 0$ at $r = R$ (no slip):

$$u = (1/4\mu)(-\partial p/\partial x)(R^2 - r^2) \quad \text{(parabolic profile)}$$



Pipe: Umax = 2 x average velocity

Plates: Umax = 1.5 x average velocity

3.3 Important results

$$u_{max}\ (\text{at } r = 0) = (1/4\mu)(-\partial p/\partial x)\, R^2 = 2 \times V_{avg}$$

$$V_{avg} = (1/8\mu)(-\partial p/\partial x)\, R^2 = u_{max}/2$$

$$Q = \pi R^4 (-\partial p/\partial x)/8\mu = \pi D^4 \Delta p/(128\,\mu L) \quad \text{(Hagen-Poiseuille)}$$

$$\Delta p = 32\,\mu V L/D^2 \;;\; h_f = 32\,\mu V L/(\rho g D^2) = 128\,\mu Q L/(\pi \rho g D^4)$$

$$\text{Wall shear stress } \tau_0 = 8\,\mu V/D = (\Delta p/L)(D/4)$$

$$\text{Darcy friction factor (laminar): } f = 64/Re$$

- Radius at which local velocity = average velocity: **r = 0.707 R** .

3.4 Solved Problem 1

Oil ($\mu = 0.1$ Pa.s, $\rho = 900$ kg/m^3) flows at 1 L/s through a 50 mm pipe, L = 100 m. Find Re, pressure drop, head loss, u_{max} and wall shear stress.

- $A = \pi/4 \times 0.05^2 = 1.9635 \times 10^{-3}$ m^2; $V = 0.001 / 1.9635 \times 10^{-3} =$ **0.509 m/s** .

$$Re = 900 \times 0.509 \times 0.05 / 0.1 = 229 \quad (< 2000, \text{ laminar})$$

$$\Delta p = 32 \times 0.1 \times 0.509 \times 100 / 0.05^2 = 65{,}190 \text{ Pa} \approx 65.2 \text{ kPa}$$

$$h_f = 65{,}190 / (900 \times 9.81) = 7.38 \text{ m of oil}$$

- $u_{max} = 2V =$ **1.02 m/s** ; $\tau_0 = 8 \times 0.1 \times 0.509 / 0.05 =$ **8.15 Pa** .
- Check with f: $h_f = f L V^2 / 2gD = (64/229) \times 100 \times 0.259 / (2 \times 9.81 \times 0.05) = 7.38$ m.

4. Laminar Flow Between Parallel Plates

4.1 Both plates fixed (plane Poiseuille flow)

- Plates separated by B; y measured from the lower plate.

$$u = (1/2\mu)(-\partial p/\partial x)(B y - y^2)$$

$$u_{max} \text{ (at } y = B/2) = (-\partial p/\partial x) B^2 / 8\mu \; ; \; V_{avg} = (2/3) u_{max}$$

$$\text{Discharge per unit width: } q = (-\partial p/\partial x) B^3 / 12\mu$$

$$\Delta p = 12 \mu V L / B^2 \; ; \; h_f = 12 \mu V L / (\rho g B^2)$$

$$\text{Shear stress at plates: } \tau_0 = (-\partial p/\partial x) B / 2 = 6 \mu V / B$$

4.2 One plate moving (Couette flow)

- Upper plate moves with U, lower fixed. With zero pressure gradient: **$u = U y / B$** (linear profile), $\tau = \mu U / B$.
- With pressure gradient (Couette + Poiseuille): **$u = U y/B + (1/2\mu)(-\partial p/\partial x)(By - y^2)$** . Flow can reverse near the fixed plate if the pressure gradient is adverse.

4.3 Solved Problem 2

Oil (μ = 0.08 Pa.s) flows between two fixed plates 10 mm apart with pressure gradient $-\partial p/\partial x$ = 2000 Pa/m. Find u_{max}, V_{avg}, discharge per metre width and wall shear stress.

$$u_{max} = 2000 \times (0.01)^2 / (8 \times 0.08) = 0.3125 \text{ m/s}$$

$$V_{avg} = (2/3) \times 0.3125 = 0.208 \text{ m/s}; \quad q = 2000 \times (0.01)^3 / (12 \times 0.08) = 0.00208 \text{ m}^3\text{/s per m}$$

$$\tau_0 = 2000 \times 0.01 / 2 = 10 \text{ Pa}$$

5. Laminar Flow Through an Annulus

5.1 Set-up

- Concentric annulus: **outer radius R** , **inner radius r0** ; flow along axis under pressure gradient. No-slip on both walls ($u = 0$ at $r = R$ and at $r = r_0$).

5.2 Results

$$u = (1/4\mu)(-\partial p/\partial x)\left[R^2 - r^2 + (R^2 - r_0^2) \ln(r/R) / \ln(R/r_0) \right]$$

$$Q = (\pi / 8\mu)(-\partial p/\partial x)\,[\,R^4 - r0^4 - (R^2 - r0^2)^2 / \ln(R/r0)\,]$$

$$\text{Mean velocity } V = Q / [\pi (R^2 - r0^2)]$$

- Maximum velocity occurs at $r = \sqrt{[(R^2 - r0^2) / (2 \ln(R/r0))]}$; zero shear at this radius.
- Applications: journal bearings, oil in annular gap of pistons, flow in concentric tube heat exchangers.

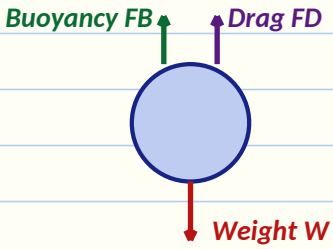
5.3 Hydraulic radius view

- For an annulus: hydraulic diameter $D_h = 4A / P = 2(R - r0)$; the limiting case $r0 \to 0$ gives the circular pipe results.

6. Stokes' Law and Falling-Sphere Viscometer

6.1 Stokes' law

For a small sphere moving slowly (Re < 0.2) through a viscous fluid, the drag force is $F_D = 3\pi\mu d V$.



At terminal velocity: W = FB + FD

Stokes drag: FD = 3 pi mu d V

Valid for Re < 0.2 (creeping flow)

$$\text{Terminal velocity: } V = g\,d^2\,(\rho s - \rho) / (18\,\mu)$$

- At terminal velocity: weight W = buoyancy FB + drag FD. Drag coefficient $C_D = 24 / Re$ for Re < 0.2.

6.2 Solved Problem 3

A steel ball (d = 2 mm, ρs = 7800) falls in glycerine (μ = 0.8 Pa.s, ρ = 1260). Find terminal velocity.

$$V = 9.81 \times (0.002)^2 \times (7800 - 1260) / (18 \times 0.8) = 0.0178 \text{ m/s}$$

- Check: $Re = 1260 \times 0.0178 \times 0.002 / 0.8 = 0.056$ (< 0.2), so Stokes' law is valid.

7. Measurement of Viscosity

7.1 Methods

Instrument	Principle	Formula
Capillary tube viscometer	Hagen-Poiseuille flow in a tube	$\mu = \pi d^4 \Delta p / (128 Q L)$
Falling sphere viscometer	Stokes' law	$\mu = g d^2 (\rho s - \rho) / (18 V)$
Rotating cylinder (Couette)	Torque on coaxial cylinders	$\mu = T \delta / (2\pi R^3 L \omega)$
Redwood / Saybolt / Engler	Time of efflux of fixed oil volume	$\nu = A t - B / t$

- **Rotating cylinder** : fluid in thin gap δ between a fixed outer and a rotating inner cylinder; measure torque T at angular speed ω.
- **Efflux viscometers** give kinematic viscosity in empirical units (Redwood seconds, Saybolt seconds); convert with the given formula.

8. Turbulent Flow in Pipes

8.1 Nature

- Velocity at a point fluctuates: $u = \bar{u} + u'$. **Reynolds stress** $= -\rho \overline{u'v'}$ (extra shear due to eddy mixing).
- Prandtl's mixing-length theory: $\tau_t = \rho l^2 (du/dy)^2$, with $l = \kappa y$ ($\kappa = 0.4$, von Karman constant).

8.2 Regions near the wall

- **Viscous (laminar) sub-layer** : very thin, thickness $\delta' = 11.6 \nu / v$; buffer zone ; **and** turbulent core .
- Shear (friction) velocity: $v = \sqrt{\tau_0 / \rho} = V \sqrt{f/8}$.

8.3 Smooth and rough pipes

Condition	Criterion (k = roughness height)
Hydraulically smooth	$k / \delta' < 0.25$ (roughness is submerged in the sub-layer)
Transition	$0.25 < k / \delta' < 6$
Hydraulically rough	$k / \delta' > 6$ (roughness projects out, causes eddies)

8.4 Velocity distribution laws

Smooth: $u / v^* = 5.75 \log_{10}(y v^* / \nu) + 5.5$

Rough: $u / v^* = 5.75 \log_{10}(y / k) + 8.5$

Velocity defect law: $(u_{max} - u) / v^* = 5.75 \log_{10}(R / y)$

- Power-law (approx): **$u / u_{max} = (y/R)^{1/n}$**, $n = 7$ for Re about 10^5 (1/7th power law); mean velocity $V \approx 0.817\, u_{max}$ for $n = 7$.

9. Resistance to Flow: Friction Factor

9.1 Darcy-Weisbach equation

$h_f = f L V^2 / (2 g D)$; $f = 8 \tau_0 / (\rho V^2)$

- (Fanning friction factor $f' = f / 4$ in some books.) Also $h_f = 4 f' L V^2 / (2gD)$.

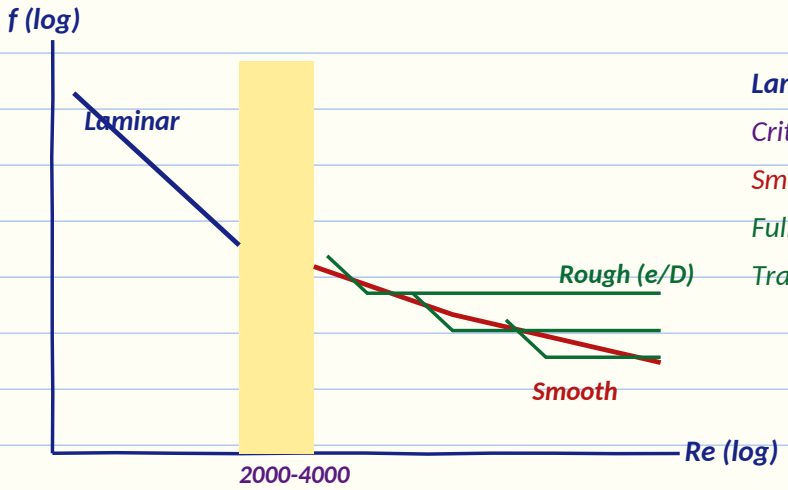
9.2 Friction factor correlations

Flow regime	Formula
Laminar (Re < 2000)	$f = 64 / Re$
Smooth turbulent, 4000 < Re < 10^5	Blasius: $f = 0.316 / Re^{0.25}$
Smooth turbulent, any Re	Prandtl: $1/\sqrt{f} = 2 \log_{10}(Re \sqrt{f}) - 0.8$
Fully rough	von Karman: $1/\sqrt{f} = 2 \log_{10}(3.7 D / k)$
Transition (smooth and rough)	Colebrook-White: $1/\sqrt{f} = -2 \log_{10}[k/(3.7D) + 2.51/(Re \sqrt{f})]$

- Explicit approximation: **Swamee-Jain**: $f = 0.25 / [\log_{10}(k/(3.7D) + 5.74/Re^{0.9})]^2$.

9.3 Moody's diagram

- A plot of **f (y-axis) versus Re (x-axis, log scale)** for various relative roughness k/D.



Laminar: f = 64 / Re

Critical zone: Re 2000 to 4000

Smooth (Blasius): f = 0.316 / Re^0.25

Fully rough: f depends on e/D only

Transition: f depends on Re and e/D

- **Laminar zone** : straight line f = 64/Re, independent of roughness. **Critical zone** (2000 to 4000): uncertain. **Smooth pipe curve** ; **transition zone** : curves of constant k/D drop and flatten; **fully rough zone** : horizontal lines, f depends only on k/D (hf V^2).
- **Usage** : compute Re and k/D, read f, then find $h_f = f\,L\,V^2 / 2gD$.

9.4 Solved Problems

Problem 4: Water ($\nu = 10^{-6}$) flows at 0.5 L/s in a 25 mm pipe. Find Re and the nature of flow.

- $V = 0.0005 / 4.909 \times 10^{-4} = 1.019$ m/s; **$Re = 1.019 \times 0.025 / 10^{-6} = 25{,}460$** (turbulent).

Problem 5: D = 0.1 m, L = 200 m, V = 2 m/s, f = 0.02. Find head loss.

$h_f = 0.02 \times 200 \times 2^2 / (2 \times 9.81 \times 0.1) = 16 / 1.962 = 8.15$ m

Problem 6: Smooth pipe at Re = 50,000. Find f (Blasius).

f = 0.316 / (50,000)^0.25 = 0.316 / 14.95 = 0.0211

Problem 7: Power to overcome friction. Q = 0.02 m^3/s water, hf = 8.15 m.

$P = \rho g Q h_f = 1000 \times 9.81 \times 0.02 \times 8.15 = 1599$ W ≈ 1.6 kW

9.5 Minor losses (other resistances in pipes)

Loss	Formula
Sudden enlargement	$h_L = (V1 - V2)^2 / 2g$
Sudden contraction	$h_L = 0.5\,V2^2 / 2g$ (approximately; Cc = 0.62)
Pipe entrance	$h_L = 0.5\,V^2 / 2g$ (sharp), 0.04 to 0.1 for rounded

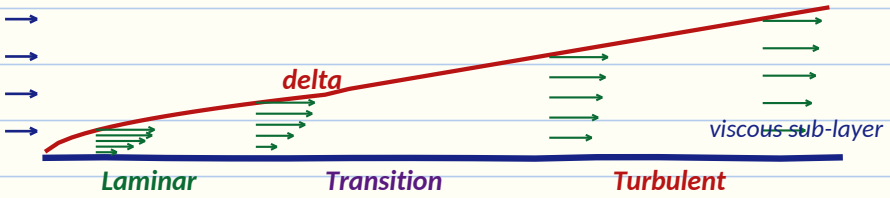
Pipe exit	$h_L = V^2 / 2g$
Bends, valves, fittings	$h_L = K V^2 / 2g$

- **Major loss** = friction along length (Darcy-Weisbach). **Minor loss** = due to change in section or direction (local disturbances). Total head loss = major + minor.

10. Introduction to Boundary Layer Theory

10.1 Concept (Prandtl, 1904)

- When a real fluid flows past a solid surface, velocity is **zero at the wall** (no-slip) and rises to the free-stream value U over a thin layer: the **boundary layer**. Outside it, flow is practically ideal (inviscid).



Boundary layer thickness delta: where u = 0.99 U

Re_x(crit) = 5 x 10^5

10.2 Thickness definitions

Term	Meaning	Formula
Boundary layer thickness δ	Distance where $u = 0.99\ U$	-
Displacement thickness δ^*	Distance wall is displaced due to velocity deficit	$\delta^* = \int (1 - u/U)\, dy$
Momentum thickness θ	Loss of momentum flux	$\theta = \int (u/U)(1 - u/U)\, dy$
Energy thickness δ^{**}	Loss of kinetic energy flux	$\int (u/U)(1 - u^2/U^2)\, dy$

10.3 Flat plate results

Quantity	Laminar ($Re_x < 5 \times 10^5$)	Turbulent
δ	$5x / \sqrt{Re_x}$ (Blasius)	$0.37x / Re_x^{0.2}$

Local Cf	0.664 / √Re_x	0.059 / Re_x^0.2
Average CD	1.328 / √Re_L	0.074 / Re_L^0.2
δ and θ	δ = 1.72 x/√Re_x; θ = 0.664 x/√Re_x	δ = δ/8 ; θ = 7δ/72

Momentum integral equation: $\tau_0 = \rho U^2 \frac{d\theta}{dx}$ (von Karman)

- Drag: $F_D = C_D \times (1/2 \rho U^2) \times A$. Transition on flat plate at $Re_x \approx 5 \times 10^5$.

10.4 Boundary layer separation

- In an **adverse pressure gradient** ($dp/dx > 0$, e.g. diverging section, rear of a cylinder), fluid near the wall loses momentum, velocity gradient at wall becomes zero and the flow **separates**, forming a wake and **form drag**.
- **Remedies**: streamlined shapes, suction or blowing, slots, rotating cylinder, turbulence promoters (dimples on golf ball).

10.5 Solved Problem 8

Air ($\nu = 1.5 \times 10^{-5}$, $\rho = 1.2$) flows at $U = 10$ m/s over a flat plate. At $x = 0.5$ m find Re_x, δ, local Cf, wall shear stress, δ and θ.

$Re_x = 10 \times 0.5 / 1.5 \times 10^{-5} = 3.33 \times 10^5$ (laminar, < 5×10^5)

- $\delta = 5 \times 0.5 / \sqrt{(3.33 \times 10^5)} = 2.5 / 577 =$ **4.33 mm** ; $C_f = 0.664 / 577 = 0.00115$.
- $\tau_0 = C_f \times 1/2 \rho U^2 = 0.00115 \times 0.5 \times 1.2 \times 100 =$ **0.069 Pa**.
- $\delta =$ **1.72 x 0.5 / 577 = 1.49 mm**; $\theta =$ **0.664 x 0.5 / 577 = 0.575 mm.**
- Drag on one side (plate 0.5 m x 1 m): $C_D = 1.328/577 = 0.0023$; $F_D = 0.0023 \times 60 \times 0.5 =$ **0.069 N**.

11. More Solved Problems

Problem 9: Laminar flow in an annulus

Oil ($\mu = 0.1$ Pa.s, $\rho = 900$) flows in an annulus with $R = 25$ mm, $r_0 = 10$ mm, under $-\partial p/\partial x = 1000$ Pa/m. Find Q, V and the radius of maximum velocity.

- $R^4 = 3.906 \times 10^{-7}$; $r_0^4 = 10^{-8}$; $(R^2 - r_0^2)^2 = (5.25 \times 10^{-4})^2 = 2.756 \times 10^{-7}$; $\ln(R/r_0) = \ln 2.5 = 0.916$.
- Bracket $= 3.906 \times 10^{-7} - 0.1 \times 10^{-7} - 3.008 \times 10^{-7} = 7.98 \times 10^{-8}$.

$Q = (\pi / 8 \times 0.1) \times 1000 \times 7.98 \times 10^{-8} = 3.13 \times 10^{-4}$ m³/s (0.313 L/s)

- Area = π (6.25 - 1) x 10^{-4} = 1.649 x 10^{-3} m^2; V = 0.19 m/s. Radius of u_max = sqrt(5.25 x 10^{-4} / 1.832) = **16.9 mm** .
- Re = 900 x 0.19 x 0.03 / 0.1 = 51, laminar (hydraulic diameter = 2(R - r0) = 30 mm).

Problem 10: Capillary tube viscometer

Tube d = 1 mm, L = 0.2 m, pressure drop 2000 Pa; 0.147 cm^3 of oil collected in 60 s. Find μ.

- Q = 0.147 x 10^{-6} / 60 = 2.45 x 10^{-9} m^3/s.

$\mu = \pi d^4 \Delta p / (128 Q L) = \pi \times 10^{-12} \times 2000 / (128 \times 2.45 \times 10^{-9} \times 0.2) = 0.1$ Pa.s

Problem 11: Rotating cylinder viscometer

R = 50 mm, L = 100 mm, gap δ = 0.5 mm, N = 60 rpm, torque T = 0.05 N.m. Find μ.

- ω = 2π x 60 / 60 = 6.283 rad/s.

$\mu = T \delta / (2\pi R^3 L \omega) = 0.05 \times 0.0005 / (2\pi \times 1.25 \times 10^{-4} \times 0.1 \times 6.283) = 0.0507$ Pa.s

Problem 12: Turbulent pipe: friction factor and sub-layer

Water (ν = 10^{-6}) in D = 0.3 m pipe, L = 500 m, k/D = 0.0005, Re = 10^6.

- Swamee-Jain: log10(1.35 x 10^{-4} + 5.74 / 10^5.4) = log10(1.58 x 10^{-4}) = - 3.80; f = 0.25 / 14.45 = **0.0173** (Moody also gives about 0.017).
- V = Re ν / D = 10^6 x 10^{-6} / 0.3 = 3.33 m/s. hf = 0.0173 x 500 x 3.33^2 / (2 x 9.81 x 0.3) = **16.3 m** .
- τ0 = ρ g hf D / 4L = 1000 x 9.81 x 16.3 x 0.3 / 2000 = 24.0 Pa; **v = sqrt(24/1000) = 0.155 m/s (check V sqrt(f/8) = 0.155).**
- δ' = 11.6 ν / v **= 11.6 x 10^{-6} / 0.155 = 0.0748 mm; k = 0.15 mm, so k/δ' = 2.0:** transition zone **(neither smooth nor fully rough).**

Problem 13: Boundary layer shape (assumed profiles)

Linear profile u/U = y/δ: **δ = ∫(1 - y/δ)dy = δ/2; θ = ∫(y/δ)(1 - y/δ)dy = δ/6; shape factor H = δ/θ = 3.**

Parabolic profile u/U = 2η - η^2 (η = y/δ): **δ = δ/3; θ = 2δ/15; H = 2.5.**

Problem 14: Turbulent boundary layer drag

Air at U = 15 m/s (ν = 1.5 x 10^{-5}) over a plate 1 m long, 1 m wide, turbulent from the leading edge.

- $Re_L = 15 \times 1 / 1.5 \times 10^{-5} = 10^6$; $Re^{0.2} = 15.85$. δ at $x = 1$ m = $0.37 / 15.85 =$ **23.3 mm** .

$CD = 0.074 / 15.85 = 0.00467$; $FD = CD \times (1/2\ \rho U^2) \times A = 0.00467 \times (0.5 \times 1.2 \times 225) \times 1 = 0.63$ N

Problem 15: Equivalent velocity profile check (1/7th power law)

Pipe R = 0.1 m, $u_{max} = 3$ m/s, $n = 7$. Velocity at $y = 0.02$ m from wall: $u = 3 \times (0.02/0.1)^{(1/7)} = 3 \times 0.795 =$ **2.39 m/s** ; mean $V = 0.817 \times 3 = 2.45$ m/s.

12. Quick Revision and Important Questions

Laminar: $\tau = \mu\ du/dy$; $f = 64/Re$; umax = 2V (pipe), 1.5V (plates). Annulus: no slip on both walls. Stokes: $F = 3\pi\mu dV$ ($Re < 0.2$). $Re < 2000$ laminar, > 4000 turbulent. Blasius $f = 0.316/Re^{0.25}$. Boundary layer: $Re_x(crit) = 5 \times 10^5$.

- Derive the velocity distribution for laminar flow through a circular pipe and obtain the Hagen-Poiseuille equation. (10 M)
- Derive expressions for laminar flow between two fixed parallel plates. (10 M)
- Explain Reynolds experiment and the transition from laminar to turbulent flow. (5 M)
- State Stokes' law; explain the falling sphere and capillary viscometers. (5 M)
- Explain Moody's diagram and the variation of friction factor for smooth and rough pipes. (10 M)
- Define displacement, momentum and energy thickness. Explain boundary layer separation. (10 M)

UNIT - II

UNIFORM FLOW IN OPEN CHANNELS

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

- Open channel flow; comparison with pipe flow
- Geometric parameters; classification of channels and flow
- Velocity distribution of channel section
- Hydraulically efficient sections: rectangular, trapezoidal, triangular
- Energy and momentum correction factors

1. Introduction to Open Channel Flow

1.1 Definition

Open channel flow is the flow of a liquid with a **free surface exposed to atmosphere** (pressure on the surface = atmospheric). Flow is driven by **gravity** (component of weight along the slope).

- Examples: rivers, canals, drains, flumes, culverts, partially filled sewers and pipes.

1.2 Comparison between open channel flow and pipe flow

Basis	Open channel flow	Pipe flow
Free surface	Present, at atmospheric pressure	Absent; pipe flows full
Driving force	Gravity (bed slope)	Pressure difference
Hydraulic gradient line	Coincides with the water surface	Above the pipe axis, a pressure head
Cross-section	Varies: shape, depth and area change	Fixed (circular)
Roughness	Varies with depth, time (silt, vegetation)	Roughly constant
Flow area	A depends on depth y	Fixed
Analysis	More difficult (extra variable y)	Simpler

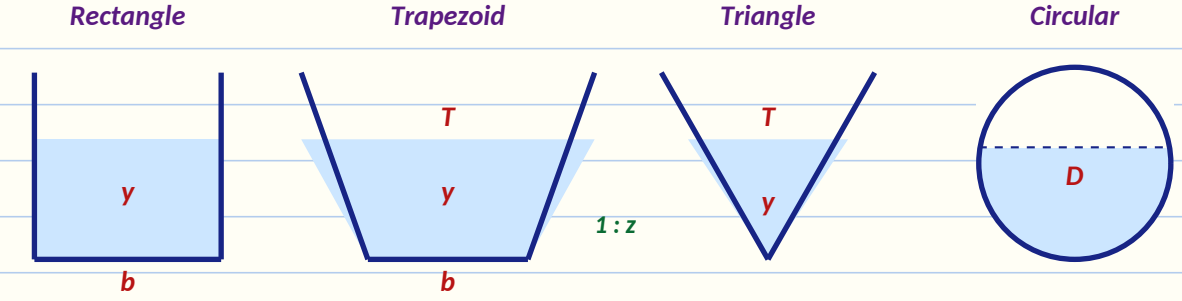
Design equation	Manning / Chezy	Darcy-Weisbach
Max velocity	Slightly below the surface	At the centre

2. Geometric Parameters of a Channel

2.1 Definitions

Term	Meaning
Depth of flow (y)	Vertical distance from the lowest point of the section to the free surface
Top width (T)	Width of the channel section at the free surface
Flow area (A)	Cross-sectional area of flow normal to the direction of flow
Wetted perimeter (P)	Length of the channel boundary in contact with water
Hydraulic radius (R)	$R = A / P$
Hydraulic depth (D)	$D = A / T$ (mean depth)
Section factor (Z)	$Z = A\sqrt{D} = A\sqrt{(A/T)}$; used for critical flow
Bed slope (S0)	Longitudinal slope of the channel bed = $\sin\theta \approx \tan\theta$
Side slope (z)	z horizontal to 1 vertical
Freeboard	Vertical distance from water surface to top of the bank

2.2 Formulas for common sections



Section	Area A	Wetted perimeter P	Top width T
Rectangle (b, y)	$b\,y$	$b + 2y$	b
Trapezoid (b, y, z)	$(b + z\,y)\,y$	$b + 2y\sqrt{(1 + z^2)}$	$b + 2\,z\,y$
Triangle (z, y)	$z\,y^2$	$2\,y\sqrt{(1 + z^2)}$	$2\,z\,y$
Circle (D, angle 2θ)	$(D^2/8)(2\theta - \sin 2\theta)$	$D\,\theta$	$D \sin\theta$

Example: Trapezoidal $b = 4$ m, $z = 1.5$, $y = 2$ m: $A = (4 + 3) \times 2 =$ **14 m²** ; $P = 4 + 2 \times 2 \times \sqrt{3.25} = 4 + 7.21 =$ **11.21 m** ; $R = 14 / 11.21 =$ **1.249 m** ; $T = 4 + 2 \times 1.5 \times 2 =$ **10 m** ; $D = 1.4$ m.

3. Classification of Open Channels

3.1 By origin

- **Natural channels** : rivers, streams, estuaries; irregular shape and roughness.
- **Artificial channels** : canals, flumes, drains, spillways, culverts; regular shape.

3.2 By cross-section and slope

- **Prismatic channel** : constant cross-section and bed slope along its length. **Non-prismatic**: varying section or slope.
- **Rigid boundary** (lined concrete) and **mobile boundary** (alluvial, erodible channel with sediment transport).
- **Shapes** : rectangular, trapezoidal, triangular, circular, parabolic, U-shaped, horse-shoe.

3.3 By bed slope

- **Mild slope** , **steep slope** , **critical slope** , **horizontal** , **adverse** (rising) slope.

4. Classification of Open Channel Flow



4.1 With respect to time

- **Steady flow** : depth, velocity do not change with time at a section ($\partial y/\partial t = 0$). **Unsteady flow** : they change (flood wave, surges).

4.2 With respect to space

- **Uniform flow** : depth, area, velocity same at every section along the channel. **Non-uniform (varied) flow** : they change along the length.
- Varied flow is of two types: **Gradually varied flow (GVF)** (depth changes slowly: backwater curve) and **Rapidly varied flow (RVF)** (depth changes abruptly over short length: hydraulic jump, flow over weir).

4.3 With respect to Reynolds number

Re = V R / ν (R = hydraulic radius)

- **Laminar**: Re < 500; **transition**: 500 to 2000 (some books: 500 to 12,500); **turbulent**: Re > 2000 (most channels are turbulent).

4.4 With respect to Froude number

Fr = V / √(g D) (D = hydraulic depth; for rectangle D = y)

- **Subcritical (tranquil)**: Fr < 1; **Critical**: Fr = 1; **Supercritical (rapid / shooting)**: Fr > 1.
- Fr is the ratio of **inertia to gravity force**. A small surface wave (celerity $c = \sqrt{g y}$) can travel **upstream only if flow is subcritical**.

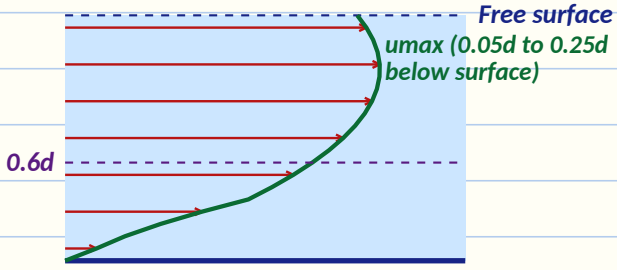
4.5 Summary

Type	Condition	Notes
Uniform steady	Sf = S0 = water surface slope	Normal depth yn
Gradually varied	y changes slowly	Backwater and drawdown curves
Rapidly varied	y changes abruptly	Hydraulic jump, drop, sluice gate

5. Velocity Distribution in a Channel Section

5.1 Pattern

- Because of friction at bed and banks and the free surface, velocity varies in both vertical and transverse directions. **Maximum velocity occurs below the free surface** (at 0.05 d to 0.25 d from the surface; deeper in narrow channels).



Mean velocity V:

V = u at 0.6 depth, or

V = (u0.2 + u0.8) / 2

V = 0.85 x surface velocity

5.2 Vertical velocity profile

- Approximately **logarithmic** (for wide channel): $u = (v/\kappa) \ln(y/y0)$, **with** $\kappa = 0.4$; **also power law** $u = u_{max} (y/d)^{(1/7)}$.

5.3 Mean velocity from measurement

Method	Mean velocity
Single point	V = u at 0.6 depth from the surface
Two-point	V = (u at 0.2d + u at 0.8d) / 2
Three-point	V = (u0.2 + 2 u0.6 + u0.8) / 4
Surface float	V = k x surface velocity, k ≈ 0.85 (0.8 to 0.95)

- **Instruments** : current meter (cup or propeller type), Pitot tube, floats, ADCP (acoustic Doppler).

5.4 Wide rectangular channel (laminar)

- $u = (g\ S0 / 2\nu)(2\ d\ y - y^2)$; the mean velocity = **$V = g\ S0\ d^2 / 3\nu$** and the maximum velocity at the surface = 1.5 V.

6. Uniform Flow Formulae

6.1 Conditions for uniform flow

- Depth, area, velocity and discharge are constant; **bed slope = water surface slope = energy slope (S0 = Sw = Sf)** .
- Force balance: **gravity component along the slope = shear resistance at wall** : $\rho\ g\ A\ S0\ L = \tau 0\ P\ L$.

$$\tau 0 = \rho\ g\ R\ S0$$

6.2 Chezy's formula (1769)

$$V = C\sqrt{(R\ S0)} \quad ; \quad Q = A\ C\sqrt{(R\ S0)}$$

- C = Chezy coefficient ($m^{1/2}/s$). Derived by equating $\tau 0 = k\ \rho\ V^2$ with $\tau 0 = \rho\ g\ R\ S0$.
- **Bazin** : C = 157.6 / (1 + m/√R); **Kutter** : empirical formula with n; **Manning**: C = R^(1/6) / n; **Powell** ; **Pavlovsky** .

6.3 Manning's formula (most used)

$$V = (1/n)\ R^{(2/3)}\ S0^{(1/2)} \quad ; \quad Q = (1/n)\ A\ R^{(2/3)}\ S0^{(1/2)}$$

- n = Manning's roughness coefficient ($s/m^{(1/3)}$).

Surface	n
Smooth cement / planed wood	0.010
Concrete, finished, brick	0.013 to 0.015
Unlined earth channel, clean	0.022 to 0.025
Natural stream with weeds	0.035 to 0.050

Rough rock / heavy vegetation	0.05 to 0.100

- **Conveyance K = (1/n) A R^(2/3)** so Q = K √S0. **Normal depth yn** = depth at which uniform flow occurs; solve A R^(2/3) = Q n / √S0 (section factor).

6.4 Solved Problems

Problem 1: Rectangular channel b = 3 m, y = 1.5 m, n = 0.015, S0 = 1/1000. Find V and Q.

- A = 4.5 m²; P = 3 + 3 = 6 m; R = 0.75 m; R^(2/3) = 0.8255.

V = (1/0.015) x 0.8255 x √0.001 = 1.74 m/s; Q = 4.5 x 1.74 = *7.83 m³/s*

Problem 2: Trapezoidal channel b = 4 m, z = 1.5, y = 2 m, n = 0.02, S0 = 0.0004. Find Q.

- A = 14 m²; P = 11.21 m; R = 1.249 m; R^(2/3) = 1.160.

V = (1/0.02) x 1.160 x 0.02 = 1.16 m/s; Q = 14 x 1.16 = *16.24 m³/s*

Problem 3: Find Chezy's C for R = 0.75 m, n = 0.015 and verify V (Problem 1).

- C = R^(1/6) / n = 0.9532 / 0.015 = 63.6; V = 63.6 x √(0.75 x 0.001) = 63.6 x 0.02739 = **1.74 m/s**. matches Manning.

Problem 4: Froude number for V = 2 m/s, y = 1.5 m (rectangular).

- Fr = 2 / √(9.81 x 1.5) = 2 / 3.836 = **0.52** (subcritical, tranquil).

Problem 5: Current-meter readings at 0.2d and 0.8d are 1.2 and 0.8 m/s; area is 5 m².

V = (1.2 + 0.8)/2 = 1.0 m/s; Q = 5 x 1.0 = 5 m³/s

7. Hydraulically Efficient Channel Sections

7.1 Concept

For a given area A, slope S0 and roughness n, the section that gives **maximum discharge** is the one with **minimum wetted perimeter P** (maximum hydraulic radius R). This is the **most economical / best hydraulic section**.

- Since Q = (1/n) A R^(2/3) S0^(1/2) = (1/n) A^(5/3) P^(-2/3) S0^(1/2). For constant A, Q is max when P is min.
- Among all shapes the **semi-circle** is the best, but it is difficult to construct; so rectangle,

trapezoid and triangle are used.

7.2 Rectangular section

- $A = b\,y$; $P = b + 2y = A/y + 2y$. Set $dP/dy = 0$: $-A/y^2 + 2 = 0$, so $A = 2y^2$.

Best rectangular section: $b = 2y$ and $R = y / 2$

- Width is twice the depth; depth of flow equals twice the hydraulic radius.

7.3 Trapezoidal section

- $A = (b + z\,y)\,y$; $P = b + 2y\sqrt{(1 + z^2)}$. Substituting $b = A/y - z\,y$, $P = A/y - z\,y + 2\,y\,\sqrt{(1 + z^2)}$. $dP/dy = 0$:

$$A = y^2\,[2\sqrt{(1 + z^2)} - z] \qquad b = 2\,y\,[\sqrt{(1 + z^2)} - z]$$

Best trapezoidal section: $R = y / 2$ (for every side slope z)

- For the best trapezoid the length of each sloping side equals half the top width (and equals b when $z = 1/\sqrt{3}$). With $z = 1/\sqrt{3}$ the section is half of a regular hexagon (side slope 60° to horizontal).
- The best side slope for a trapezoid: $z = 1/\sqrt{3} = 0.577$ (60°); then $b = 2y/\sqrt{3} = 1.155\,y$, $A = \sqrt{3}\,y^2 = 1.732\,y^2$, $P = 2\sqrt{3}\,y = 3.464\,y$.
- A semi-circle drawn with centre at the water-surface mid-point and radius y touches all three sides (bed and two banks).

7.4 Triangular section

- $A = z\,y^2$; for a given A, $P = 2\,y\,\sqrt{(1 + z^2)} = 2\,\sqrt{(A)}\,\sqrt{((1 + z^2)/z)}$. Minimum when $d/dz[(1 + z^2)/z] = 0$, so $z = 1$.

Best triangular section: $z = 1$ (side slopes 45°, vertex angle 90°); $R = y / (2\sqrt{2}) = 0.354\,y$; $A = y^2$

7.5 Circular section

- Max discharge when flow depth $y = 0.95\,D$ (θ about 154°); max velocity when $y = 0.81\,D$. At $y = D$ (full), Q is slightly lower than max, hence an enclosed pipe is not "best" when full.

7.6 Comparison for the same area $A = 8\ m^2$ (best sections)

Section	Depth y	Wetted perimeter P	R
Rectangle ($b = 2y$)	2.00 m	8.00 m	1.00 m
Trapezoid ($z = 0.577$)	2.15 m	7.44 m	1.075 m
Triangle ($z = 1$)	2.83 m	8.00 m	1.00 m
Semi-circle (reference)	$r = 2.26$ m	7.09 m	1.13 m

- The half-hexagon trapezoid has the least perimeter among practical sections.

7.7 Solved Problems

Problem 6: Design the best rectangular section to carry Q = 10 m^3/s, n = 0.015, S0 = 1/2500.

- b = 2y; A = $2y^2$; R = y/2. Q = (1/n) A R^(2/3) √S0 = 66.67 x $2y^2$ x 0.63 y^(2/3) x 0.02 = 1.68 y^(8/3).

1.68 y^(8/3) = 10 y^(8/3) = 5.95 *y = 1.95 m*, b = 3.90 m; V = 10 / 7.62 = 1.31 m/s

Problem 7: Best trapezoidal section for A = 8 m^2 (z = 1/√3).

- A = 1.732 y^2 = 8, so y = 2.149 m; b = 1.155 y = 2.48 m; P = 3.464 x 2.149 = 7.44 m; R = y/2 = 1.07 m.

Problem 8: Triangular section, best form, Q = 2 m^3/s, n = 0.02, S0 = 0.001. Find y.

- A = y^2; R = 0.354 y; Q = (1/0.02) x y^2 x (0.354 y)^(2/3) x 0.0316 = 50 x 0.0316 x 0.5 y^(8/3) = 0.79 y^(8/3).

0.79 y^(8/3) = 2 y^(8/3) = 2.53 y = 1.41 m (top width T = 2y = 2.82 m)

8. Energy and Momentum Correction Factors

8.1 Why needed

- Velocity is not uniform over the cross-section, so the **true kinetic energy flux and momentumflux** differ from those computed using the mean velocity V.

8.2 Kinetic energy correction factor (Coriolis coefficient α)

$$\alpha = (1 / A V^3) \int u^3 \, dA \quad (\alpha \geq 1)$$

- Kinetic energy per unit weight = $\alpha V^2 / 2g$. Specific energy: $E = y + \alpha V^2 / 2g$.

8.3 Momentum correction factor (Boussinesq coefficient β)

$$\beta = (1 / A V^2) \int u^2 \, dA \quad (\beta \geq 1)$$

- Momentum flux = $\beta \rho Q V$. Momentum equation: F = ρ Q (β2 V2 - β1 V1).
- Typical values: **α = 1.03 to 1.36**, **β = 1.01 to 1.12**; α > β always (α - 1 ≈ 3(β - 1)). Uniform velocity: α = β = 1.
- Laminar flow in a pipe: α = 2, β = 4/3; laminar between plates: α = 54/35, β = 6/5.

8.4 Power-law profile

u = u_max η^m (η = y/d): α = (1 + m)³ / (1 + 3m) ; β = (1 + m)² / (1 + 2m)

$m = 1/7$: $\alpha = (8/7)^3 / (10/7) = 1.045$; $\beta = (8/7)^2 / (9/7) = 1.016$.

8.5 Solved Problem 9

The section is divided into 4 equal areas where mean velocities are 1.0, 1.5, 2.0 and 1.5 m/s. Find α and β.

- Mean $V = (1.0 + 1.5 + 2.0 + 1.5)/4 = 1.5$ m/s.

$$\alpha = \Sigma u^3 / (n V^3) = (1 + 3.375 + 8 + 3.375) / (4 \times 3.375) = 15.75 / 13.5 = *1.167*$$

$$\beta = \Sigma u^2 / (n V^2) = (1 + 2.25 + 4 + 2.25) / (4 \times 2.25) = 9.5 / 9 = *1.056*$$

9. Design Aspects of Channels

9.1 Permissible velocity

- Velocity must be high enough to **prevent silting** (minimum velocity, about 0.6 to 0.9 m/s) and low enough to **prevent scouring** (maximum permissible velocity, depends on boundary material).

Boundary material	Approx. permissible velocity (m/s)
Fine sand	0.45
Sandy loam	0.55
Silt loam / ordinary firm loam	0.6 to 0.75
Stiff clay	1.1
Fine gravel	1.5
Hard pan, lined concrete	1.8 to 3 or more

9.2 Side slopes (z) and freeboard

- Typical side slopes (horizontal : vertical): **rock 0 to 0.25; stiff clay 0.5 to 1; firm earth 1 to 1.5; loose earth / sandy soil 2 to 3.** Freeboard is typically 0.3 to 1.0 m in canals.

9.3 Tractive force (shear) approach

$$\tau_0 = \gamma R S_0 \quad \text{(average boundary shear stress)}$$

- On the sloping sides the maximum shear is about 0.76 γ y S0; on the bed it is about γ y S0. Channel is stable if τ0 < permissible tractive force of the soil.

9.4 Composite roughness and compound sections

- **Composite roughness** (Horton / Einstein): different n_i over parts P_i of the perimeter:

$$n_e = [\ \Sigma P_i n_i^{1.5} / P\ ]^{(2/3)}$$

- **Compound section** (river with flood plains): divide into sub-areas, compute Q for each (using its own A, R and n) and add.

9.5 Circular channels partly full

- Flow depth ratio y/D: **Q is max at y/D = 0.95** and **V is max at y/D = 0.81** . At half-full, R = D/4 (same as full) so V is the same as full pipe; **Q(half) = 0.5 Q(full)**.

10. More Solved Problems

Problem 9: Normal depth by trial

Rectangular channel b = 4 m, Q = 8 m³/s, n = 0.015, S0 = 0.0004. Find yn.

- Required A R^(2/3) = Q n / √S0 = 8 x 0.015 / 0.02 = **6.0** .

y (m)	A	P	R	A R^(2/3)
1.5	6.0	7.0	0.857	5.41
1.6	6.4	7.2	0.889	5.92
1.62	6.48	7.24	0.895	6.02

Normal depth yn ≈ 1.617 m (V = 8 / 6.47 = 1.24 m/s)

Problem 10: Composite roughness

Rectangular channel b = 4 m, y = 2 m. Bed n = 0.015; side walls n = 0.025. Find equivalent n.

- P1 (bed) = 4 m; P2 (two sides) = 4 m; P = 8 m. n_e = [(4 x 0.015^1.5 + 4 x 0.025^1.5) / 8]^(2/3) = (0.002895)^(2/3).

n_e = *0.0203*

Problem 11: Circular culvert half full

D = 1.2 m, depth y = 0.6 m (half full), n = 0.013, S0 = 0.002. Find V and Q.

- A = π D² / 8 = 0.5655 m²; P = π D / 2 = 1.885 m; R = D/4 = 0.3 m; R^(2/3) = 0.448.

V = (1/0.013) x 0.448 x √0.002 = 1.54 m/s; Q = 0.5655 x 1.54 = *0.87 m³/s* (half of the full-pipe 1.74 m³/s)

Problem 12: Required bed slope

Trapezoidal canal b = 5 m, z = 1, y = 2 m, n = 0.025 must carry Q = 20 m³/s. Find S0.

- A = 14 m²; P = 5 + 2 x 2 x 1.414 = 10.66 m; R = 1.314 m; R^(2/3) = 1.20; V = 20/14 =

1.43 m/s.

$S0 = (V\,n / R^{(2/3)})^2 = (1.43 \times 0.025 / 1.20)^2 = 8.87 \times 10^{-4}$ (about 1 in 1130)

Problem 13: Discharge of best trapezoidal section

$y = 1.5$ m, $z = 1/\sqrt{3}$, $n = 0.02$, $S0 = 0.0005$ (best section). $A = 1.732\,y^2 = 3.90$ m²; $R = y/2 = 0.75$ m.

$Q = (1/0.02) \times 3.90 \times 0.8255 \times 0.02236 = *3.60\ m^3/s*$

Problem 14: Boundary shear stress

Using Problem 2: $R = 1.249$ m, $S0 = 0.0004$.

$\tau 0 = 1000 \times 9.81 \times 1.249 \times 0.0004 = 4.9\ N/m^2$ (compare with the soil's permissible tractive force)

Problem 15: Reynolds and Froude numbers

Rectangular channel, $b = 2$ m, $y = 0.5$ m, $V = 3$ m/s, $\nu = 10^{-6}$ m²/s.

- $R = A/P = 1.0/3.0 = 0.333$ m; $Re = V R / \nu = 3 \times 0.333 / 10^{-6} = 10^6$ (turbulent). $Fr = 3 / \sqrt{(9.81 \times 0.5)} = 1.35$ (> 1, supercritical).

11. Derivations and Discharge Estimation

11.1 Derivation of Chezy's formula

- Consider uniform flow in a length L. **Forces**: weight component $W \sin\theta = \rho g A L S0$ (driving); **resisting force** $= \tau 0\, P\, L$.
- For uniform flow they balance: $\rho g A L S0 = \tau 0 P L$, so $\tau 0 = \rho g R S0$.
- Experiments show resistance varies as the square of velocity: $\tau 0 = k \rho V^2$.

$k \rho V^2 = \rho g R S0 \quad V = \sqrt{(g/k)}\,\sqrt{(R S0)} = C\sqrt{(R S0)} \quad$ where $C = \sqrt{(g/k)}$

11.2 Velocity-area method of discharge measurement

- Divide the cross-section into strips of width w; measure depth d and mean velocity v in each strip with a current meter. $Q = \Sigma (w\,d\,v)$.

Example: 4 strips of width 2 m; depths 0.5, 1.0, 1.5, 1.0 m; mean velocities 0.4, 0.7, 0.9, 0.6 m/s.

$Q = 2 \times (0.5 \times 0.4 + 1.0 \times 0.7 + 1.5 \times 0.9 + 1.0 \times 0.6) = 2 \times 2.85 = 5.7\ m^3/s$

11.3 Slope-area method (floods)

- Use two sections of a reach: Manning with the average conveyance, $Q = K\sqrt{Sf}$, where K

= (1/n)(A R^(2/3)) average of sections, and Sf found from energy equation between the sections.

11.4 Chezy vs Manning

Basis	Chezy	Manning
Formula	V = C √(R S0)	V = (1/n) R^(2/3) S0^(1/2)
Coefficient	C varies with R and roughness	n depends mainly on roughness only
Use	Pipes and channels, theory	Standard in practice for channels
Relation	C = R^(1/6) / n	n = R^(1/6) / C

12. Quick Revision and Important Questions

Open channel: free surface, gravity-driven. R = A/P; D = A/T; Fr = V/√(gD). Manning: V = (1/n)R^(2/3)S^(1/2). Best sections: rectangle b = 2y; trapezoid half-hexagon, R = y/2; triangle z = 1. $\alpha = \int u^3 dA/(AV^3)$; $\beta = \int u^2 dA/(AV^2)$.

- Compare open channel flow with pipe flow. Classify open channel flow with examples. (5 M)
- Explain the velocity distribution in an open channel and the methods of finding mean velocity. (5 M)
- Derive Chezy's formula and state Manning's formula. (5 M)
- Derive the conditions for the most economical (i) rectangular (ii) trapezoidal (iii) triangular section. (10 M)
- Define energy and momentum correction factors; derive them. (5 M)
- Problems: discharge for given section, design of best section, Froude number classification.

UNIT - III

NON-UNIFORM FLOW IN OPEN CHANNELS

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

- Specific energy, critical flow, discharge curve
- Specific force, specific depth, critical depth
- Measurement of discharge and velocity
- Gradually varied flow; dynamic equation of GVF
- Hydraulic jump and classification; elements, characteristics; energy dissipation

1. Specific Energy

1.1 Definition

Specific energy E is the **energy per unit weight of water measured with respect to the channel bed (as datum)**.

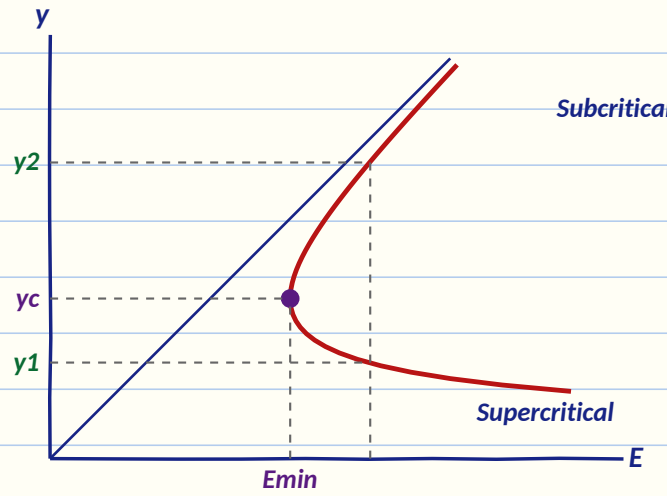
$$E = y + \alpha V^2 / 2g = y + Q^2 / (2 g A^2) \quad (\text{take } \alpha = 1)$$

- Total energy $H = z + y + V^2/2g$ changes along the channel because of the bed slope, but E changes only because of friction (E decreases in the flow direction when $S0 = 0$).

1.2 Rectangular channel (discharge per unit width q = Q/b)

$$E = y + q^2 / (2 g y^2)$$

- For a constant q, E is a **cubic** in y: for each $E > Emin$ there are **two positive roots (alternate depths)** y1 (supercritical, small depth, high velocity) and y2 (subcritical, large depth, low velocity), and one negative root.



$E = y + V^2 / 2g = y + q^2 / (2 g y^2)$

Two alternate depths (y1, y2) for one E; one at E = Emin

$y_c = (q^2 / g)^{1/3}$, $E_c = 1.5\, y_c$

1.3 Critical flow (minimum specific energy)

- Differentiate E with respect to y and put dE/dy = 0 (constant Q): $1 - Q^2 T / (g A^3) = 0$.

  $Q^2 T / (g A^3) = 1$ i.e. $Fr = 1$ or $A^3 / T = Q^2 / g$ (section factor $Z = Q/\sqrt{g}$)

- Rectangular channel results:

  $y_c = (q^2 / g)^{1/3}$; $V_c = \sqrt{(g\, y_c)}$; $E_c = 1.5\, y_c$; $(V_c^2/2g = y_c/2)$

- **Critical depth yc** depends only on discharge and channel shape, not on slope or roughness.

Channel	Critical depth relation	Ec
Rectangular	$y_c = (q^2 / g)^{1/3}$	1.5 yc
Triangular (side slope z)	$y_c = (2 Q^2 / (g z^2))^{1/5}$	1.25 yc
Trapezoidal / circular	By trial from $A^3/T = Q^2/g$	by calculation
Parabolic	-	1.333 yc

1.4 Classification by Froude number

State	Condition	Depth
Subcritical	Fr < 1	y > yc (V < Vc)
Critical	Fr = 1	y = yc
Supercritical	Fr > 1	y < yc (V > Vc)

1.5 Specific depth and critical slope

- **Specific depth** : the depth of flow corresponding to a given specific energy; the two **alternate depths** are the two specific depths for a given E.
- **Critical slope Sc** : bed slope at which uniform flow occurs at critical depth. For wide

rectangular channel (Manning): $S_c = g\,n^2 / y_c^{1/3}$.

- Mild slope ($S_0 < S_c$): $y_n > y_c$, subcritical uniform flow. Steep slope ($S_0 > S_c$): $y_n < y_c$. Critical slope: $y_n = y_c$.

2. Discharge Curve (Constant Specific Energy)

- For a fixed E: $q = y\sqrt{2g(E - y)}$. Plot q against y: q = 0 at y = 0 and at y = E, and passes through a **maximum**.

$$dq/dy = 0 \quad y = 2E/3 = y_c ; \quad q_{max} = \sqrt{g}\,(2E/3)^{3/2}$$

- So for a given specific energy the discharge is **maximum at critical depth**, and for a given discharge the specific energy is **minimum at critical depth**. A channel transition (hump, contraction) works on this principle.

Choking and effect of a hump or contraction

- For a rise in the bed (hump) of height Δz in subcritical flow, depth falls; E at the hump = $E1 - \Delta z$. If Δz becomes larger than E1 - Ec, flow is **choked**: upstream depth rises (backwater) so that critical flow occurs over the hump.
- Similarly, reducing the width b increases q; maximum q is reached when flow is critical at the contraction (**choking**).

Solved problems

Problem 1: Rectangular channel with $q = 2.5\ m^2/s$. Find yc, Vc and Ec.

$y_c = (2.5^2 / 9.81)^{1/3} = (0.637)^{1/3} = 0.86$ m; $V_c = 2.5 / 0.86 = 2.91$ m/s; $E_c = 1.5 \times 0.86 = 1.29$ m

Problem 2: Alternate depth: $q = 2\ m^2/s$, $y_1 = 0.5$ m. Find y2.

- $E1 = 0.5 + 4 / (19.62 \times 0.25) = 0.5 + 0.8155 = 1.3155$ m. Solve $y + 0.2039 / y^2 = 1.3155$ by trial: y = 1.15 gives 1.304; y = 1.17 gives 1.319.

$y_2 \approx 1.166$ m (subcritical, since $y_c = 0.742$ m)

Problem 3: Maximum discharge for E = 2 m (rectangular, per metre width).

$y_c = 2E/3 = 1.333$ m; $q_{max} = \sqrt{9.81} \times 1.333^{1.5} = 3.132 \times 1.540 = 4.82\ m^2/s$

Problem 4: Triangular channel z = 1.5, $Q = 1.5\ m^3/s$. Find yc.

$y_c = [2 \times 2.25 / (9.81 \times 2.25)]^{1/5} = (0.2039)^{0.2} = 0.727$ m (check: $Q^2 T / g A^3 = 1$)

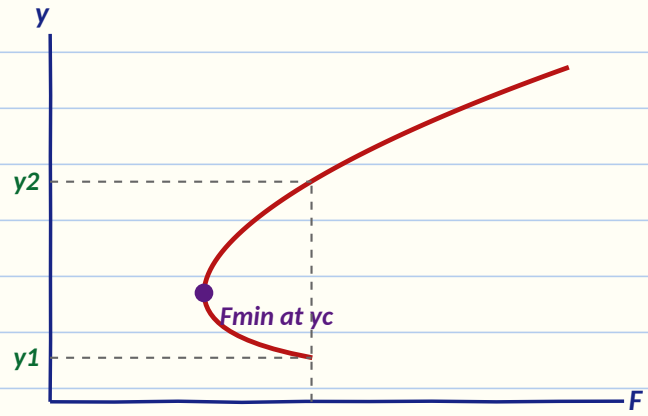
3. Specific Force (Momentum Function)

3.1 Definition

Specific force F = **(momentum flux + hydrostatic pressure force) per unit weight of water**, between two sections with no external force.

$$F = Q^2 / (g A) + A \bar{y} \quad (\bar{y} = \text{depth of centroid of area below water surface})$$

- Rectangular channel: $F = q^2 / (g y) + y^2 / 2$ (per unit width).



Specific force (momentum function):

$F = Q^2/(g A) + A \cdot \bar{y}$

Rectangular: $F = q^2/(g y) + y^2/2$

y1 and y2 = conjugate depths (equal F, hydraulic jump)

- The curve has a **minimum at critical depth**; for a given F there are two depths (**sequent or conjugate depths**) y1 and y2: same specific force but different specific energy. This is the principle of the hydraulic jump.

3.2 Specific energy vs specific force

Basis	Specific energy	Specific force
Based on	Energy equation (no loss assumed)	Momentum equation (no external force)
Two depths	Alternate depths	Conjugate (sequent) depths
Used for	Smooth transitions, gates, humps	Hydraulic jump, force on structures
Minimum	At critical depth	At critical depth

- In a hydraulic jump, F is conserved but E is lost.

Check using the jump data of Problem 8: q = 4, y1 = 0.5: F1 = 16 / (9.81 x 0.5) + 0.125 = 3.387; y2 = 2.316: F2 = 16 / (9.81 x 2.316) + 2.682 = 3.386 (equal).

4. Measurement of Discharge and Velocity

4.1 Velocity measurement

- **Current meter** (cup or propeller): n revolutions per second, v = a + b n (calibrated). **Pitot tube**, floats (surface, sub-surface, rod floats), ADCP, dye / salt dilution method.
- Mean velocity from one-point (0.6d) or two-point (0.2d, 0.8d) readings (Unit II).

4.2 Discharge measurement

Device	Discharge formula
Rectangular sharp-crested weir	$Q = (2/3)\, Cd\, \sqrt{(2g)}\, L\, H^{(3/2)}$
Francis formula (with end contractions)	$Q = 1.84\,(L - 0.1\, n\, H)\, H^{(3/2)}$
Triangular (V) notch	$Q = (8/15)\, Cd\, \sqrt{(2g)}\, \tan(\theta/2)\, H^{(5/2)}$
Broad-crested weir	$Q = 1.705\, Cd\, L\, H^{(3/2)}$ (max flow, critical depth on crest)
Venturi flume / Parshall flume	Throat at critical flow: $Q = Cd\, (2/3)^{(3/2)}\, \sqrt{g}\, b\, H^{(3/2)}$
Sluice (underflow) gate	$Q = Cd\, b\, a\, \sqrt{(2\, g\, y1)}$
Velocity-area method	$Q = \Sigma$ (width x depth x mean velocity)

- **Critical-depth meters** (weirs, flumes, Venturi flume) are reliable: Q depends on only upstream head.

Problem 5: Rectangular weir L = 2 m, H = 0.3 m, Cd = 0.62. Q = (2/3) x 0.62 x 4.429 x 2 x 0.3^1.5 = **0.60 m^3/s**.

Problem 6: 90° V-notch, H = 0.2 m, Cd = 0.6. Q = (8/15) x 0.6 x 4.429 x tan 45° x 0.2^2.5 = **0.0254 m^3/s**.

5. Gradually Varied Flow (GVF)

5.1 Meaning and assumptions

- GVF is **steady non-uniform flow in which the depth changes gradually along the channel length**, so that streamlines are nearly parallel and pressure is hydrostatic. Examples: backwater above a dam or weir, drawdown near a free overfall.
- Assumptions: **steady flow**; **prismatic channel**; hydrostatic pressure; small bed slope; friction slope Sf given by Manning / Chezy as in uniform flow with the local depth; α =

1.

5.2 Dynamic equation of GVF (derivation)

- Total head $H = z + y + V^2/2g$. Differentiate with respect to x: **$dH/dx = -Sf$** (energy slope), $dz/dx = -S0$.
- $dH/dx = -S0 + dy/dx + d(V^2/2g)/dx = -Sf$. Since $V = Q/A$, $d(V^2/2g)/dx = -(Q^2 T / g A^3)$ $dy/dx = -Fr^2\, dy/dx$.

$$dy/dx = (S0 - Sf) / (1 - Fr^2) \quad \text{(dynamic equation)}$$

- Using Manning for a wide rectangular channel with normal depth yn and critical depth yc:

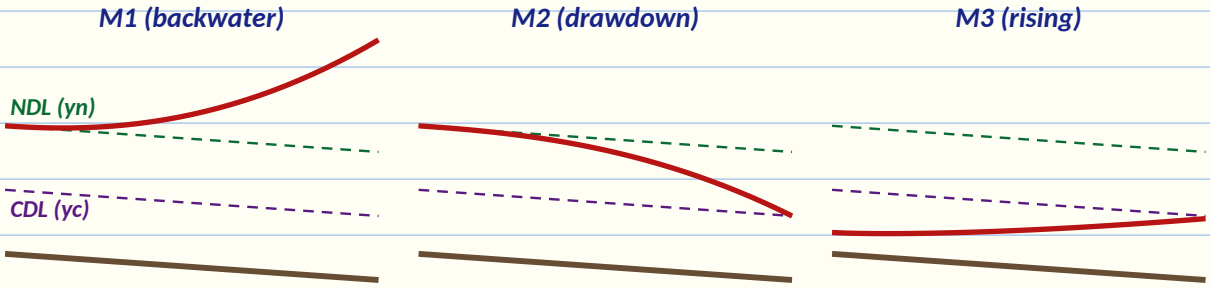
$$dy/dx = S0\,[\,1 - (yn / y)^{(10/3)}\,] / [\,1 - (yc / y)^3\,]$$

- **Numerator** : positive when $y > yn$ ($S0 > Sf$); **denominator** : positive when $y > yc$ (subcritical).

5.3 Classification of water surface profiles

- Zone 1: y above both lines; zone 2: between normal and critical; zone 3: below both. Slopes: Mild (M), Steep (S), Critical (C), Horizontal (H), Adverse (A).

Profile	Condition	dy/dx	Nature	Example
M1	$y > yn > yc$	+	Backwater (rising)	Dam on mild slope
M2	$yn > y > yc$	-	Drawdown (falling)	Approach to a free overfall
M3	$yn > yc > y$	+	Rising, ends in jump	Below sluice gate on mild slope
S1	$y > yc > yn$	+	Backwater	Mild-to-steep, behind a jump
S2	$yc > y > yn$	-	Drawdown	Break from mild to steep
S3	$yc > yn > y$	+	Rising	Below gate on steep slope
C1, C3	$y > yc$ (or $y < yc$)	+	Rising	Critical slope
H2, H3	$yc > y$ or $y > yc$	- or +	No normal depth ($S0 = 0$)	Horizontal bed
A2, A3	$y > yc$, $y < yc$	- / +	No normal depth ($S0 < 0$)	Adverse slope



Mild slope: yn > yc. M1: y > yn; M2: yc < y < yn; M3: y < yc

- Rule: **profiles in zone 2 are drawdown curves; in zones 1 and 3 they rise in the flow direction.** Near yn the profile approaches normal depth asymptotically; near yc it meets critical depth at a steep angle (dy/dx → infinity).

5.4 Computation methods

- **Direct step method** : $\Delta x = (E2 - E1) / (S0 - Sf)$; step depth-wise. Easiest for prismatic channels.
- **Standard step method** : for natural channels; trial and error on depth at the next section.
- **Graphical integration** , **numerical (Runge-Kutta)** , and **Bresse's method** (wide rectangular channels).

Direct step: $\Delta x = (E2 - E1) / (S0 - Sf)$, $Sf = (Sf1 + Sf2)/2$ with $Sf = n^2 V^2 / R^{(4/3)}$

5.5 Solved Problem 7 (backwater)

Rectangular channel b = 10 m, Q = 20 m^3/s (q = 2 m^2/s), n = 0.02, S0 = 0.001. A dam raises depth to 1.50 m. Find the distance upstream at which the depth is 1.40 m.

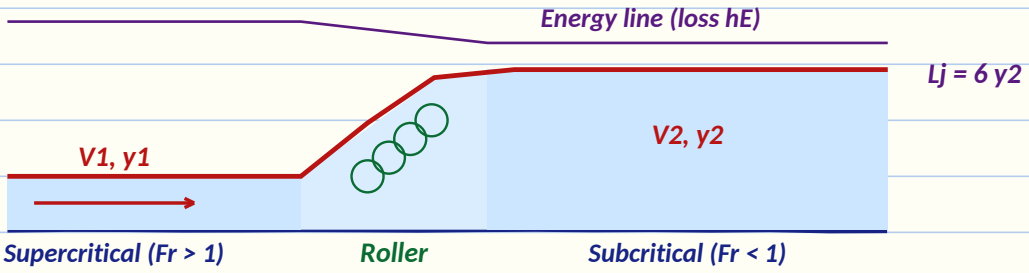
- yc = (4/9.81)^(1/3) = 0.742 m; wide channel yn ≈ (q n/√S0)^(3/5) = 1.15 m, so slope is mild and 1.5 > yn > yc: **M1 curve** .
- Section 1 (y = 1.5): V = 1.333; E1 = 1.5 + 0.0906 = 1.5906 m; R = 15/13 = 1.154 m; Sf1 = 0.0004 x 1.778 / 1.154^(4/3) = 5.88 x 10^{-4}.
- Section 2 (y = 1.4): V = 1.429; E2 = 1.4 + 0.1040 = 1.5040 m; R = 14/12.8 = 1.094 m; Sf2 = 7.24 x 10^{-4}.

Δx = (1.5040 - 1.5906) / (0.001 - 0.000656) = - 0.0866 / 0.000344 = - 252 m (about 252 m upstream)

6. Hydraulic Jump (Rapidly Varied Flow)

6.1 Definition

A hydraulic jump is the **sudden rise of water surface when a supercritical (rapid) stream changes into a subcritical (tranquil) stream** ; it is accompanied by **turbulence, a surface roller, air entrainment and large energy loss** .



- It occurs when the downstream depth (tailwater) is enough to sustain the **conjugate depth** y2 of the incoming supercritical flow, e.g. below a spillway, sluice gate or at a change from steep to mild slope.

6.2 Momentum equation (rectangular channel)

- Neglect bed friction and weight component; the **change in momentum flux = difference in hydrostatic forces** (per unit width):

$$\rho\, q\,(V2 - V1) = (1/2)\,\gamma\, y1^2 - (1/2)\,\gamma\, y2^2$$

$$y2 / y1 = (1/2)\,[\,\sqrt{(1 + 8\,Fr1^2)} - 1\,] \quad \text{and} \quad y1 / y2 = (1/2)\,[\,\sqrt{(1 + 8\,Fr2^2)} - 1\,]$$

- $Fr1 = V1 / \sqrt{(g\, y1)}$ must be > 1; (y1, y2) are **conjugate (sequent) depths**.

6.3 Elements (characteristics) of the jump

Element	Value / Formula
Height of jump	$hj = y2 - y1$
Energy loss	$hL = E1 - E2 = (y2 - y1)^3 / (4\, y1\, y2)$
Efficiency	$\eta = E2 / E1$ (ratio of downstream to upstream specific energy)
Length of jump	$Lj \approx 6\, y2$ (about 5 to 7 y2); roller length $Lr \approx 4$ to 5 y2
Power dissipated	$P = \gamma\, Q\, hL$
Relative energy loss	$hL / E1$ increases with Fr1 (about 45 to 70% at Fr1 = 4 to 9)

6.4 Classification of jumps (USBR, by upstream Froude number)

Fr1	Type	Description	Energy loss
1.0 - 1.7	Undular	Standing waves, no true jump	< 5%

1.7 - 2.5	Weak	Smooth rise, small rollers	5 - 15%
2.5 - 4.5	Oscillating	Unstable pulsating jet; waves travel downstream	15 - 45%
4.5 - 9.0	Steady (stable)	Well-balanced, stable, best for dissipation	45 - 70%
> 9.0	Strong (choppy)	Rough, high-velocity jet, intermittent slugs	70 - 85%

- Other classifications: by tailwater (**free, repelled, submerged jump**), jump on sloping floor, and **forced jump** (with baffles or sill).

6.5 Applications / uses

- **Dissipation of energy** below dams, spillways, weirs, sluices (prevents scour); **raising water level** downstream of a measuring flume; **mixing of chemicals** (water treatment); **aeration** of water; removal of air pockets in pipelines; maintaining high water level in irrigation channels.

6.6 Energy dissipation: stilling basins

- A **stilling basin** is a paved apron structure at the foot of a spillway in which the hydraulic jump is **contained** so that the high-energy flow is converted to low-energy flow before entering the river.
- USBR basins: **Type I** (plain, Fr1 < 2.5); **Type II** (chute blocks + dentated end sill, Fr1 > 4.5, V1 > 15 m/s); **Type III** (chute blocks + baffle blocks + end sill, V1 < 18 m/s, small structures); **Type IV** (for oscillating jump, Fr1 between 2.5 and 4.5).
- Accessories: **chute blocks** (break the jet), **baffle (friction) blocks** , **end sill** (reduces length of basin and guides flow), **tailwater depth** at least y2. **Roller bucket** and **ski-jump bucket** for high dams.

6.7 Solved Problem 8

A rectangular channel carries $q = 4\ m^2/s$ at $y1 = 0.5$ m. Find Fr1, y2, energy loss, efficiency, length and power dissipated per metre width.

- V1 = 4 / 0.5 = 8 m/s; **Fr1 = 8 / √(9.81 x 0.5) = 3.61** (oscillating jump).

y2 = (0.5 / 2) [√(1 + 8 x 13.05) - 1] = 0.25 x (10.265 - 1) = *2.316 m*

hL = (2.316 - 0.5)³ / (4 x 0.5 x 2.316) = 5.989 / 4.632 = *1.29 m*

- E1 = 0.5 + 64 / 19.62 = 3.762 m; V2 = 4 / 2.316 = 1.727; E2 = 2.316 + 0.152 = 2.468 m; check E1 - E2 = 1.294 m.
- Efficiency E2/E1 = 65.6%; Lj = 6 x 2.316 = **13.9 m** ; P = 9810 x 4 x 1.29 = **50.6 kW per metre width** .

Problem 9: Conjugate depth from y2

A jump forms with y2 = 3 m, V2 = 1.0 m/s. Fr2 = 1 / √(9.81 x 3) = 0.184; y1 = (3/2)[√(1 + 8 x 0.0339) - 1] = 1.5 x (1.1273 - 1) = **0.19 m** .

7. Applications and More Solved Problems

7.1 Control sections

- A **control section** has a definite depth-discharge relation (usually critical depth). Examples: **weir crest** , **free overfall** (end depth ≈ 0.715 yc), **break in grade from mild to steep** , **sluice gate** , **Venturi flume throat** .
- Subcritical flow is controlled from **downstream** ; supercritical flow is controlled from **upstream**. Computations of GVF start at a control section and proceed upstream (subcritical) or downstream (supercritical).

7.2 Flow over a hump (channel transition)

Problem 10: q = 2 m^2/s, upstream depth 1.2 m. Find the depth over a 0.15 m hump and the maximum hump height without choking.

- V1 = 1.667; E1 = 1.2 + 0.1416 = 1.3416 m; yc = 0.742 m; Ec = 1.113 m.

  Max hump (no choking) = E1 - Ec = 1.3416 - 1.113 = 0.229 m
- For Δz = 0.15 m: E2 = 1.3416 - 0.15 = 1.1916 m; y + 0.2039 / y^2 = 1.1916 gives **y ≈ 0.98 m** (subcritical, so surface falls over the hump).

7.3 Contraction in width

Problem 11: Q = 8 m^3/s, b1 = 4 m, y1 = 1.5 m. Find minimum width at which choking begins.

- V1 = 1.333; E1 = 1.5906 m. Maximum q at E1: q_max = √g (2E/3)^(3/2) = 3.132 x (1.0604)^1.5 = 3.42 m^2/s.

  b_min = Q / q_max = 8 / 3.42 = 2.34 m (narrower than this causes choking and backwater)

7.4 Sluice gate followed by a jump

Problem 12: Upstream depth 3 m, depth after the gate (vena contracta) y1 = 0.4 m. Find q, Fr1, y2 and energy loss (neglect approach velocity).

- 3 = 0.4 + $V1^2$ / 2g, so V1 = √(2 x 9.81 x 2.6) = 7.14 m/s; q = 0.4 x 7.14 = **2.86 m^2/s** .

- Fr1 = 7.14 / √(9.81 x 0.4) = 3.6; y2 = 0.2 (√(1 + 8 x 13) - 1) = **1.85 m** .

hL = (1.85 - 0.4)³ / (4 x 0.4 x 1.85) = 3.04 / 2.96 = 1.03 m

7.5 Direct step method: second step

- Continue Problem 7 ($q = 2\ m^2/s$, $b = 10\ m$, $n = 0.02$, $S0 = 0.001$):

y (m)	V	E (m)	R (m)	Sf x 10^{-4}	Mean Sf	Δx (m)	
1.50	1.333	1.5906	1.154	5.88	-	-	
1.40	1.429	1.5040	1.094	7.24	6.56	- 252	
1.30	1.538	1.4206	1.032	9.08	8.16	- 454	

- Total distance from the dam where y = 1.30 m = 252 + 454 = **706 m upstream** . Steps become longer as the curve approaches yn (1.15 m) asymptotically.

7.6 Jump relations for various Froude numbers (rectangular channel)

Fr1	y2 / y1	Energy loss hL / E1	Jump type
2	2.37	9%	Weak
3	3.77	26%	Oscillating
5	6.59	49%	Steady
9	12.24	70%	Strong

- **Rule** : a steady jump (Fr1 = 4.5 to 9) is the most efficient and most stable energy dissipator; avoid the oscillating range (Fr1 = 2.5 to 4.5) in design.

7.7 Difference between GVF and RVF

Basis	Gradually varied flow	Rapidly varied flow
Depth change	Slow, over long distance	Abrupt, over a short distance
Pressure	Hydrostatic	Non-hydrostatic, curved streamlines
Losses	Friction governs	Local turbulence governs (energy loss)
Examples	Backwater, drawdown	Hydraulic jump, flow at gate, over a weir

8. Quick Revision and Important Questions

$E = y + V^2/2g$. $yc = (q^2/g)^{(1/3)}$; $Ec = 1.5\ yc$; $Fr = 1$. q_{max} at $y = 2E/3$. $F = Q^2/gA + A\bar{y}$. $dy/dx = (S0 - Sf)/(1 - Fr^2)$. Jump: $y2/y1 = 0.5[\sqrt{(1 + 8Fr1^2)} - 1]$; $hL = (y2 - y1)^3/(4y1y2)$; $Lj = 6y2$.

- Define specific energy; draw the specific energy curve and derive the conditions for critical flow. (10 M)
- Define specific force and show that the specific force is minimum at critical depth. (5 M)
- Derive the dynamic equation of gradually varied flow. Explain the classification of GVF profiles. (10 M)
- Derive conjugate depth relations for a hydraulic jump in a rectangular channel. Obtain energy loss. (10 M)
- Classify hydraulic jumps. Describe the uses of the jump and stilling basins as energy dissipators. (5 M)
- Problems: critical depth, alternate depths, GVF by direct step method, jump height, loss and power.

UNIT - IV

IMPACT OF JETS & HYDRAULIC TURBINES

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

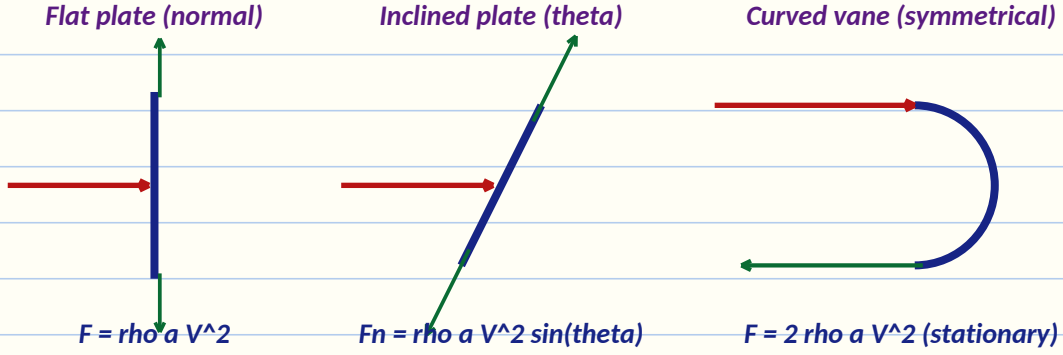
- Hydrodynamic force of jets on stationary and moving flat, inclined and curved vanes
- Velocity triangles at inlet and outlet; work done and efficiency
- Hydraulic turbines: classification; Pelton wheel and its design
- Francis turbine and its design; efficiency; draft tube theory
- Characteristic curves of hydraulic turbines; cavitation: causes and effects

1. Impact of Jets

1.1 Principle

A jet of water striking a vane exerts a **force** equal to the **rate of change of momentum** of the jet in that direction (Newton's second law): $F = \rho Q (V_{in} - V_{out})$.

- Jet velocity V (m/s), area $a = \pi d^2/4$, discharge $Q = a V$, mass flow rate $\rho a V$.



1.2 Stationary vanes

Vane	Force on vane (direction of jet)
Flat plate normal to jet	$F = \rho a V^2$
Inclined plate (angle θ to jet)	Normal force $Fn = \rho a V^2 \sin\theta$; $Fx = \rho a V^2 \sin^2\theta$; $Fy = \rho a V^2 \sin\theta \cos\theta$

Symmetrical curved vane (jet at centre, tip angle θ)	$F = \rho a V^2 (1 + \cos\theta)$; semi-circular vane ($\theta = 0$): $F = 2 \rho a V^2$
Unsymmetrical curved vane (jet at one tip, tip angles θ, φ)	$Fx = \rho a V^2 (\cos\theta + \cos\varphi)$; $Fy = \rho a V^2 (\sin\theta - \sin\varphi)$

- A curved vane gives **more force** than a flat plate because the jet is turned (deflected); a semi-circular vane gives twice the force.

1.3 Moving vanes (plates moving with velocity u in the direction of the jet)

- **Relative velocity** of jet = (V - u). Only the **relative** mass flow striking the single moving plate is $\rho a (V - u)$.

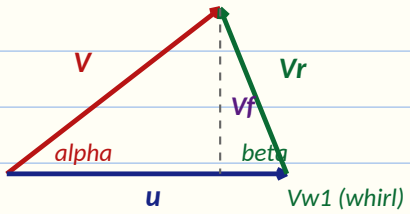
Case	Force	Work done / s	Efficiency
Single moving flat plate (normal)	$\rho a (V - u)^2$	$\rho a (V - u)^2 u$	$2u (V - u)^2 / V^3$; max 29.6% at $u = V/3$
Series of flat plates (wheel)	$\rho a V (V - u)$	$\rho a V (V - u) u$	$2u (V - u) / V^2$; max 50% at $u = V/2$
Single curved vane (moving)	$\rho a (V - u)^2 (1 + \cos\theta)$	F x u	$2u (V - u)^2 (1 + \cos\theta) / V^3$
Series of curved vanes (wheel)	$\rho a V (V - u)(1 + \cos\theta)$	F x u	$2u (V - u)(1 + \cos\theta) / V^2$; max $(1 + \cos\theta)/2$ at $u = V/2$

- **Series of vanes** (many vanes on a wheel) intercept the whole jet, so the entire mass flow $\rho a V$ is used (not $\rho a (V - u)$). For semi-circular vanes ($\theta = 0$) the maximum efficiency is 100%.
- Jet KE per second = $(1/2) \rho a V^3$ (input power). Efficiency = work done per second / KE per second.

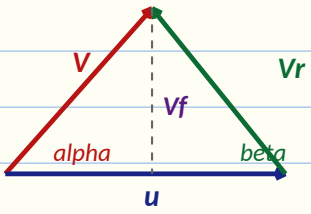
1.4 Velocity triangles

- At **inlet** and **outlet** of a moving vane draw: V (absolute velocity of jet), u (vane velocity), Vr (relative velocity), Vw (whirl component, along u), Vf (flow component, perpendicular to u), α (angle of V with u), β (angle of Vr).

Inlet triangle



Outlet triangle



$V = u + Vr$ (vector sum); $Vr1 = V1 - u1$ (for tangential entry); $Vr2 = k\ Vr1$ (k = friction factor, $k \le 1$)

Work done per second = $\rho Q (Vw1 \pm Vw2)\ u$ (Euler's equation, per unit mass: $u (Vw1 \pm Vw2)$)

- Use + sign when Vw2 is in the opposite direction to u (blade angle at outlet is acute / obtuse as per triangle), and - sign when Vw2 is in the same direction as u.

1.5 Solved Problems

Problem 1: Jet d = 50 mm, V = 20 m/s strikes (a) a stationary flat plate normally; (b) a plate moving at 8 m/s in the direction of the jet; (c) a series of plates at 10 m/s. Find force, power and efficiency.

- $a = 1.9635 \times 10^{-3}\ m^2$; jet KE rate = $0.5 \times 1000 \times 1.9635 \times 10^{-3} \times 20^3 = 7854$ W.

(a) $F = \rho a V^2 = 1000 \times 1.9635 \times 10^{-3} \times 400 = 785.4$ N

(b) $F = \rho a (V - u)^2 = 1.9635 \times 144 = 282.7$ N; power = $282.7 \times 8 = 2262$ W; $\eta = 2262 / 7854 = 28.8\%$

(c) $F = \rho a V (V - u) = 1.9635 \times 20 \times 10 = 392.7$ N; power = 3927 W; $\eta = 50\%$

Problem 2: Same jet on an inclined stationary plate at $\theta = 30°$ to the jet axis.

- $Fn = \rho a V^2 \sin\theta = 785.4 \times 0.5 = 392.7$ N; Fx (along jet) $= Fn \sin\theta = 196.4$ N; $Fy = Fn \cos\theta = 340.1$ N.

Problem 3: Same jet on a symmetrical curved vane (tip angle $\theta = 20°$): stationary and moving ($u = 5$ m/s, single vane).

- Stationary: $F = \rho a V^2 (1 + \cos 20°) = 785.4 \times 1.9397 = 1523.5$ N.
- Moving: $F = \rho a (V - u)^2 (1 + \cos 20°) = 1.9635 \times 225 \times 1.9397 = 857$ N; power $= 857 \times 5 = 4285$ W.

2. Hydraulic Turbines: Classification

2.1 Definition and heads

- A **hydraulic turbine** converts the hydraulic energy of water (head) into mechanical energy (shaft power), which is converted into electrical energy by a generator.
- **Gross head Hg** = difference of head race and tail race levels. **Net (effective) head H = Hg - hf** (hf = head loss in penstock).

2.2 Classification

Basis	Types
Action of water	**Impulse** (Pelton): only kinetic energy, atmospheric pressure at runner. **Reaction** (Francis, Kaplan): both kinetic and pressure energy; runner is full
Head	High (> 250 m): Pelton. Medium (60 to 250 m): Francis. Low (< 60 m): Kaplan, propeller
Direction of flow	Tangential (Pelton); radial inward (old Francis); mixed (modern Francis); axial (Kaplan, propeller)
Specific speed	Low (Pelton, Ns up to 60); medium (Francis, 60 to 400); high (Kaplan, > 300)
Shaft position	Horizontal or vertical

2.3 Impulse vs Reaction turbine

Basis	Impulse (Pelton)	Reaction (Francis / Kaplan)
Energy at inlet	Only kinetic (after the nozzle)	Kinetic + pressure
Pressure over runner	Atmospheric	Falls from inlet to outlet
Casing	Not essential (only to prevent splashing)	Essential, water-tight
Admission	Partial (jets)	Full circumference
Draft tube	Not required	Required
Head and discharge	High head, low discharge	Medium / low head, large discharge

2.4 Efficiencies and power

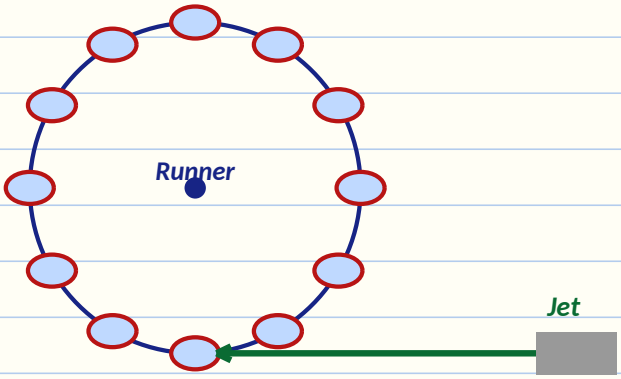
Water power WP = ρ g Q H ; Shaft power SP = 2π N T / 60

Hydraulic efficiency ηh = Runner power / Water power ; Mechanical efficiency ηm = Shaft power / Runner power

Volumetric efficiency ηv = (Q - leakage) / Q ; Overall efficiency ηo = SP / WP = ηh x ηm x ηv

3. Pelton Wheel (Impulse Turbine)

3.1 Main parts



Impulse turbine: high head, low discharge. Works at atmospheric pressure.

Bucket is double-hemispherical with a splitter in the centre.

- **Penstock** (pipe from reservoir); **Nozzle with spear** (needle valve regulates discharge); **Runner with buckets** (double hemispherical cups with splitter); **Casing** (prevents splash, guides water to tailrace); **Breaking jet** (stops the runner quickly after shut-down); **Deflector** (protects penstock from water hammer).

3.2 Working

- Water at high head is converted into a high-velocity jet in the nozzle; jet hits the splitter at the centre of the bucket, divides into two halves, and is deflected through about **165°** (not 180° to avoid striking the next bucket). Almost all KE is given to the wheel; water leaves with very small velocity.

3.3 Velocity triangle and work done

- Inlet: $V1$ = jet velocity (tangential), $u1 = u2 = u$, $Vr1 = V1 - u$, $Vw1 = V1$. Outlet: $Vr2 = k\,Vr1$, deflection angle $(180° - \varphi)$, $\varphi = 15°$, $Vw2 = Vr2 \cos\varphi - u$.

$$\text{Force} = \rho Q (V1 - u)(1 + k\cos\varphi) \;;\; \text{Power} = \rho Q (V1 - u)(1 + k\cos\varphi)\, u$$

$$\eta h = 2 (V1 - u)(1 + k\cos\varphi)\, u / V1^2 \;;\; \text{maximum at } u = V1/2:\ \eta h,max = (1 + k\cos\varphi)/2$$

- If $k = 1$ and $\varphi = 0$, $\eta h,max = 100\%$ (ideal). In practice $u \approx 0.45\ V1$ (speed ratio about 0.43 to 0.48) with $\eta 0 = 80$ to 90%.

3.4 Design of a Pelton wheel

Item	Formula / typical value
Jet velocity	$V1 = Cv\sqrt{(2\,g\,H)}$, $Cv = 0.97$ to 0.99
Bucket (wheel) velocity	$u = \varphi s\sqrt{(2\,g\,H)}$, speed ratio $\varphi s = 0.43$ to 0.48
Jet diameter	$Q = n\,(\pi/4)\,d^2\,V1$ (n = number of jets, up to 6)
Wheel (pitch circle) diameter D	$D = 60\,u / (\pi\,N)$; jet ratio $m = D/d = 10$ to 15 (up to 20)
Number of buckets	$Z = 15 + D/(2\,d) = 15 + m/2$ (typically 17 to 24)

Bucket dimensions	Width = 5 d, depth = 1.2 d, axial width about 3d to 4d
Specific speed	Ns = N √P / H^(5/4); 10 to 35 (one jet), up to 60 (multi-jet)
Regulation	Spear valve (main), jet deflector (quick closure)

3.5 Solved Problem 4 (Pelton design)

Net head 300 m, Cv = 0.98, speed ratio 0.45, jet diameter 100 mm, wheel diameter 1.4 m, deflection 165° (φ = 15°), k = 0.9. Find V1, u, Q, N, power, efficiencies, specific speed.

- V1 = 0.98 √(2 x 9.81 x 300) = 0.98 x 76.72 = **75.2 m/s** ; u = 0.45 x 76.72 = **34.5 m/s** .
- Q = (π/4)(0.1)² x 75.2 = **0.591 m³/s** ; N = 60 u / π D = 60 x 34.5 / (π x 1.4) = **471 rpm** ; D/d = 14, Z = 15 + 7 = 22 buckets.
- Vr1 = 75.2 - 34.5 = 40.7; Vr2 = 0.9 x 40.7 = 36.6; Vr2 cos 15° = 35.35.

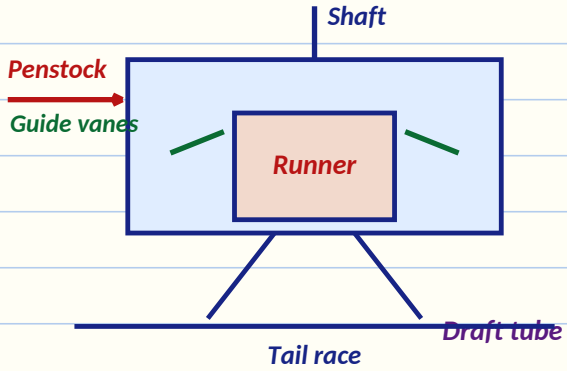
Power = 1000 x 0.591 x (40.7 + 35.35) x 34.5 = 1.55 x 10⁶ W = *1550 kW*

- Jet KE rate = 0.5 x 1000 x 0.591 x 75.2² = 1.67 MW, so **ηh = 1.55 / 1.67 = 92.8%** (based on jet KE).
- Water power = 1000 x 9.81 x 0.591 x 300 = 1.74 MW, so efficiency based on water power = 89.2% (includes nozzle loss).

Ns = N √P / H^1.25 = 471 x √1550 / 300^1.25 = 471 x 39.4 / 1249 = 14.9 (single-jet Pelton range)

4. Francis Turbine (Reaction, Mixed Flow)

4.1 Main parts



Reaction turbine (mixed flow):
medium head, medium discharge.
Water enters radially inward, leaves axially into draft tube.
Runner flows full of water.

- **Spiral (scroll) casing** with decreasing area (constant velocity); **stay ring / stay vanes** ; **guide vanes (wicket gates)** adjustable to regulate discharge and give correct entry angle; **runner** with 16 to 24 curved blades; **draft tube** (diverging) to the tailrace.

4.2 Working

- Water enters the casing at high pressure, passes through the stay and guide vanes, enters the runner **radially inwards**, gives energy to the blades while its pressure falls, and leaves **axially** (mixed flow) into the draft tube.

4.3 Velocity triangles and work done

- Inlet: V1, Vw1, Vf1 (radial), $u1 = \pi D1 N / 60$, inlet blade angle θ, guide vane angle α. Outlet: $u2 = \pi D2 N/60$, Vw2, Vf2.

Work done per second = ρ Q (Vw1 u1 ± Vw2 u2) ; Euler head He = (Vw1 u1 ± Vw2 u2) / g

For radial discharge (Vw2 = 0): P = ρ Q Vw1 u1 ; ηh = Vw1 u1 / (g H)

Flow rate Q = π D1 B1 Vf1 = π D2 B2 Vf2 (reduce by blade thickness factor if given)

- $\tan\theta = Vf1 / (Vw1 - u1)$ (for Vw1 > u1, θ < 90°); $\tan\alpha = Vf1 / Vw1$.

4.4 Design parameters

- Speed ratio **Ku = u1 / √(2gH) = 0.6 to 0.9** ; flow ratio **Kf = Vf1 / √(2gH) = 0.15 to 0.30** ; **B1/D1 = 0.1 to 0.4** ; D2 depends on specific speed (D2/D1 less than 1 for slow runners); **Ns = 60 to 300** .

4.5 Solved Problem 5 (Francis)

H = 60 m, N = 300 rpm, D1 = 1.2 m, B1 = 0.15 m, ηh = 90%, Kf = 0.2, radial discharge. Find guide vane angle, inlet blade angle, Q and power.

- $u1 = \pi \times 1.2 \times 300 / 60 =$ **18.85 m/s** . $\eta h = Vw1\ u1 / gH$, so $Vw1 = 0.9 \times 9.81 \times 60 / 18.85$ = **28.1 m/s** .
- $Vf1 = 0.2\sqrt{(2 \times 9.81 \times 60)} = 0.2 \times 34.31 =$ **6.86 m/s** .

$\alpha = \tan^{-1}(6.86 / 28.1) = 13.7°$; $\theta = \tan^{-1}[6.86 / (28.1 - 18.85)] = \tan^{-1} 0.742 = 36.6°$

$Q = \pi D1 B1 Vf1 = \pi \times 1.2 \times 0.15 \times 6.86 = 3.88\ m^3/s$

Power = ρ Q Vw1 u1 = 1000 x 3.88 x 28.1 x 18.85 = 2.06 MW (water power 2.28 MW, ηh = 90%)

- $Ns = 300\sqrt{2055} / 60^{\wedge}1.25 = 300 \times 45.3 / 167 =$ **81** (medium-speed Francis).

5. Kaplan and Propeller Turbines (Axial Flow)

- Water flows **axially** through the runner; the runner has **3 to 8 blades** on a hub (boss). **Propeller** has fixed blades; **Kaplan** has **adjustable blades** (and adjustable guide vanes: **double regulation**), so efficiency stays high over wide load variation.
- Suited for **low head (5 to 70 m) and large discharge** , Ns = 300 to 1000. Boss diameter Db = 0.35 to 0.6 D0.

$Q = (\pi/4)(D0^2 - Db^2)\, Vf1$; $u1 = u2 = \pi\, D0\, N / 60$ (vane velocity same at inlet and outlet)

- Inlet and outlet velocity triangles are drawn at the mean diameter (Dm); blades twisted along their length because u varies with radius.

Turbine	Head (m)	Ns	Flow	Regulation
Pelton	250 to 2000	10 to 60	Tangential	Spear valve, deflector
Francis	60 to 250	60 to 300	Mixed / radial	Guide vanes
Kaplan	5 to 70	300 to 1000	Axial	Guide vanes + runner blades

6. Draft Tube

6.1 Purpose

- A **gradually expanding tube** connecting the runner outlet to the tailrace; it (i) allows the runner to be placed **above the tailwater level** for easy inspection without loss of head, and (ii) **converts the large kinetic energy at the runner outlet into pressure**, thereby creating a **suction (negative) pressure** at the runner outlet which increases the effective head.

6.2 Theory

- Apply the energy equation between the runner outlet (2) and tail race surface, with runner outlet at height Hs above tailwater and V3 = exit velocity from tube:

$$p2/\gamma = pa/\gamma - Hs - (V2^2 - V3^2)/2g + hf$$

- The pressure at runner outlet is **below atmospheric**; Hs must be limited (cavitation). Draft tube efficiency:

$$\eta d = [\, V2^2/2g - V3^2/2g - hf \,] / (\, V2^2/2g \,) \quad \text{(typically 70 to 90\%)}$$

6.3 Types

- **Conical (straight divergent)**: cone angle 8° to 10° (to prevent flow separation), ηd about 90%; **simple elbow**; **elbow with varying cross-section** (rectangular at outlet); **moody spreading tube** (with a central cone-type pier).

Problem 6: Hs = 4 m, V2 = 6 m/s, V3 = 1.5 m/s, hf = 0.2 m, pa/γ = 10.33 m. $V2^2/2g$ = 1.835 m; $V3^2/2g$ = 0.115 m.

$p2/\gamma = 10.33 - 4 - (1.835 - 0.115) + 0.2 = 4.81$ m (absolute), i.e. 5.5 m below atmospheric; $\eta d = (1.835 - 0.115 - 0.2) / 1.835 = 82.8\%$

7. Unit Quantities, Governing and Characteristic Curves

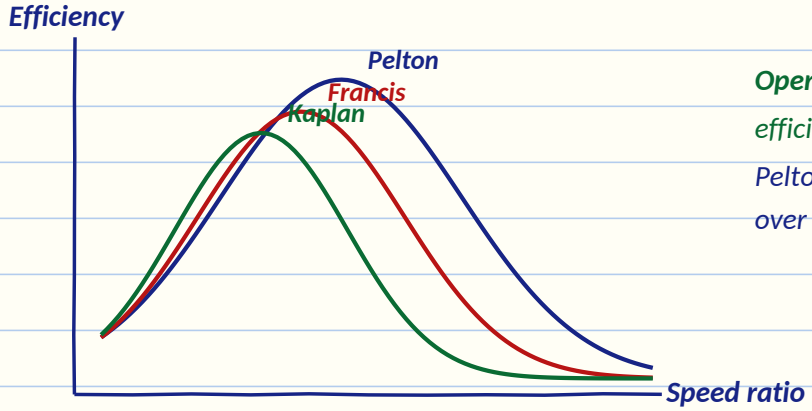
7.1 Unit quantities (head of 1 m)

Unit speed Nu = N / √H ; Unit discharge Qu = Q / √H ; Unit power Pu = P / H^(3/2)

- Used to predict the performance of the **same turbine under different heads** and of model-prototype.

Specific speed Ns = N √P / H^(5/4) (speed of a geometrically similar turbine developing 1 kW under 1 m head)

7.2 Characteristic curves



Operating (constant head) curves:
efficiency vs speed ratio / load.
Pelton efficiency stays high over a wide range of load.

Curve	Plot	Constant
Main (constant head)	Q, P and η against N	Head H
Operating (constant speed)	P, η, Q against load (gate opening)	Speed N
Constant efficiency (Muschel / iso-efficiency)	Contours of η on N-Q plane	η

- **Pelton** : efficiency stays high from 30% to 100% of load; **Francis** : drops at part load; **Kaplan** : nearly flat (adjustable blades); **propeller**: falls sharply at part load.

7.3 Governing of turbines

- **Governing** = keeping speed constant when load changes. Pelton: spear valve (and deflector); Francis: guide vane opening; Kaplan: guide vanes and blade pitch. Controlled by an oil pressure servomotor with a **centrifugal governor**.

8. Cavitation

8.1 Meaning

Cavitation is the **formation, growth and collapse of vapour bubbles** in a flowing liquid at places where the local absolute pressure falls to the **vapour pressure**. When the bubbles move to a high-pressure zone they collapse violently, producing shock waves.

8.2 Causes

- Low pressure at the **runner outlet, blade suction side or draft tube inlet**; **high runner speed or high velocity**; **excessive suction height Hs** (runner too high above tailwater); **high water temperature** (higher vapour pressure); high elevation (lower atmospheric pressure); rough surfaces and sudden changes in flow direction.

8.3 Effects

- **Pitting and erosion** of blade surface (pressures of thousands of atmospheres on the collapse of bubbles); **noise and vibration**; **drop in efficiency and output**; **damage to the runner and draft tube**; sudden failure in extreme cases.

8.4 Thoma's cavitation parameter

$\sigma = (Ha - Hv - Hs) / H$ (Ha = atmospheric head, Hv = vapour pressure head, Hs = runner height above tailwater)

- Cavitation is avoided if $\sigma > \sigma c$ (critical value from tests; larger for higher specific speed). Hence **safe** $Hs = Ha - Hv - \sigma c\, H$.

Problem 7: H = 60 m, σc = 0.10, Ha = 10.3 m, Hv = 0.3 m. $Hs = 10.3 - 0.3 - 0.10 \times 60 =$ **4.0 m** (the runner must not be placed more than 4 m above the tailwater; or set it lower).

8.5 Prevention

- Install the runner **below** (or at a limited height above) tailwater level; use a **draft tube** with correct angle; **smooth, polished** surfaces and good blade profile; **cavitation-resistant materials** (stainless steel, hard-faced welding, bronze); **air admission** (aeration) to the flow; operate within recommended load range.

9. More Solved Problems and Concepts

Problem 8: Series of curved vanes with outlet angle

A jet (d = 75 mm, V = 30 m/s) strikes a series of curved vanes moving at u = 15 m/s. The vane outlet angle is φ = 30° (smooth vanes, k = 1). Find force, power and efficiency.

- $a = 4.418 \times 10^{-3}$ m²; jet KE rate = $0.5 \times 1000 \times 4.418 \times 10^{-3} \times 30^3 = 59.6$ kW.

$F = \rho a V (V - u)(1 + \cos\varphi) = 1000 \times 4.418 \times 10^{-3} \times 30 \times 15 \times 1.866 = 3710$ N

Power = 3710 x 15 = 55.7 kW; $\eta = 2u(V - u)(1 + \cos\varphi)/V^2 = 2 \times 15 \times 15 \times 1.866 / 900 = 93.3\%$

Problem 9: Design of a multi-jet Pelton

P = 6000 kW, H = 500 m, N = 500 rpm, ηo = 85%, Cv = 0.98, speed ratio 0.46, two jets. Find Q, jet diameter, wheel diameter, number of buckets and Ns.

- $Q = P / (\rho g H \eta o) = 6 \times 10^6 / (1000 \times 9.81 \times 500 \times 0.85) =$ **1.44 m³/s**.
- $V1 = 0.98 \sqrt{(2 \times 9.81 \times 500)} = 97.1$ m/s; Q per jet = 0.72 m³/s, $a = 0.72/97.1 = 7.41 \times 10^{-3}$ m², so **d = 97 mm**.

u = 0.46 x 99.05 = 45.6 m/s; D = 60 u / π N = 60 x 45.6 / (π x 500) = 1.74 m (D/d = 17.9)

Z = 15 + D/2d = 15 + 9 = 24 buckets; Ns = 500 √6000 / 500^1.25 = 16.4 (per wheel)

Problem 10: Kaplan turbine

H = 25 m, N = 150 rpm, runner D0 = 3.5 m, hub Db = 1.4 m, Kf = 0.5, ηh = 90%. Find Q, blade angle, guide vane angle, power and Ns.

- $Vf1 = 0.5 \sqrt{(2 \times 9.81 \times 25)} = 11.07$ m/s; $Q = (\pi/4)(3.5^2 - 1.4^2) \times 11.07 = 8.08 \times 11.07 =$ **89.5 m³/s**.
- Mean diameter Dm = 2.45 m; $u = \pi \times 2.45 \times 150 / 60 =$ **19.24 m/s**; $Vw1 = \eta h\, g\, H / u = 0.9 \times 9.81 \times 25 / 19.24 = 11.47$ m/s.

Guide vane angle $\alpha = \tan^{-1}(11.07/11.47) = 44°$; inlet blade angle $\theta = 180° - \tan^{-1}[11.07 / (19.24 - 11.47)] = 180° - 54.9° = 125°$

Power = ρ Q Vw1 u = 1000 x 89.5 x 11.47 x 19.24 = 19.7 MW; Ns = 150 √19740 / 25^1.25 = 377 (Kaplan range)

Problem 11: Unit quantities

A turbine at N = 400 rpm under H = 80 m gives Q = 3 m³/s and P = 2000 kW. Find its performance under H = 60 m.

- $Nu = 400/\sqrt{80} = 44.7$; $Qu = 3/8.944 = 0.335$; $Pu = 2000 / 80^{1.5} = 2.795$.

At 60 m: N = 44.7 x 7.746 = 346 rpm; Q = 0.335 x 7.746 = 2.60 m³/s; P = 2.795 x 60^1.5 = 1299 kW

Runaway speed

- Speed attained by the turbine when the **external load is thrown off** and the governor fails (gate fully open): Pelton about 1.8 to 1.9 N; Francis 2.0 to 2.2 N; Kaplan 2.5 to 3.0 N. The generator and runner must be designed to withstand it.

Selection of a turbine (summary)

Site condition	Choice
Very high head (> 300 m), small discharge	Pelton

Medium head (60 to 250 m), moderate discharge	Francis
Low head (< 60 m), large discharge	Kaplan / propeller
Wide variation in load or head	Kaplan (adjustable blades) or Pelton

Hydro-electric power plant layout



- Components: dam and reservoir, intake with trash rack, **surge tank** (relieves water hammer), penstock, power house with turbine-generator, draft tube and tail race.

10. Quick Revision and Important Questions

Flat plate: $\rho a V^2$. Curved: $\rho a V^2(1 + \cos\theta)$. Moving wheel: $\eta max = 50\%$ at $u = V/2$ (flat); $(1 + \cos\theta)/2$ (curved). Pelton $\eta h = 2u(V1 - u)(1 + k \cos\varphi)/V1^2$. Francis $\eta h = Vw1u1/gH$. $Ns = N\sqrt{P}/H^{(5/4)}$. $\sigma = (Ha - Hv - Hs)/H$.

- Derive the force exerted by a jet on a moving curved vane. Show that the max efficiency for a series of vanes is when $u = V/2$. (10 M)
- Classify hydraulic turbines. Differentiate impulse and reaction turbines. (5 M)
- Explain the working of a Pelton wheel with sketch. Derive its efficiency and explain its design. (10 M)
- Explain the construction and working of a Francis turbine; derive the work done and efficiency. (10 M)
- What is a draft tube? Explain its types and theory. (5 M)
- Explain the characteristic curves and unit quantities. What is specific speed? (5 M)
- What is cavitation? Explain its causes, effects and prevention in hydraulic turbines. (5 M)

UNIT - V

PUMPS (CENTRIFUGAL PUMPS)

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

- Working principle of a centrifugal pump; work done by impeller
- Heads, losses and efficiencies; minimum starting speed; priming
- Specific speed; limitation of suction lift; NPSH
- Performance and characteristic curves; cavitation effects
- Multistage centrifugal pumps; troubles and remedies

1. Introduction and Working Principle

1.1 Pumps: classification

- A **pump** converts mechanical energy (from a motor / engine) into **hydraulic energy** (head) of the liquid.

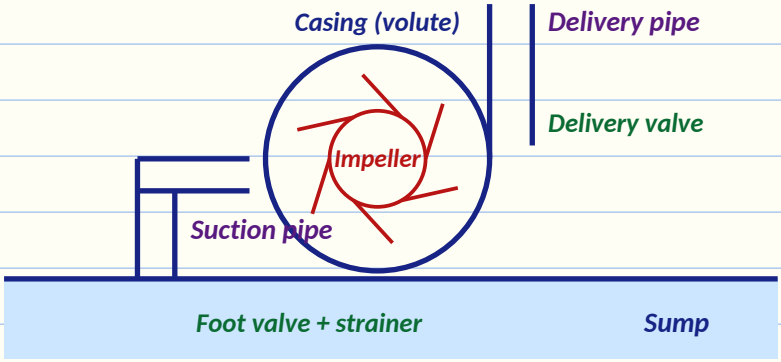
Type	Examples
Rotodynamic (kinetic)	Centrifugal (radial), mixed flow, axial flow pumps
Positive displacement	Reciprocating (piston, plunger), rotary (gear, vane, screw) pumps

- Centrifugal pumps are the most common: simple, compact, smooth flow, suitable for large discharge and moderate head.

1.2 Principle

A centrifugal pump works on the principle of **forced vortex flow**: when a mass of liquid is rotated by an impeller, a **centrifugal head** is created, so the liquid is thrown outward from the centre with increased pressure and velocity.

- Water is sucked into the **eye** of the impeller (low pressure at the centre), whirled outward by the vanes, collected in the casing (volute) where its kinetic energy is converted to pressure, and delivered through the delivery pipe.



Hs = suction head

Hd = delivery head

Hm = Hs + Hd + losses

1.3 Main parts

- **Impeller**: wheel with backward-curved vanes; types: open, semi-open, closed (shrouded); single or double suction.
- **Casing** : air-tight passage; types: **volute casing** (spiral, area increases gradually), **vortex casing** (circular chamber between impeller and volute), **casing with guide vanes (diffuser)** in turbine pumps.
- **Suction pipe** with **foot valve** (non-return) and **strainer** ; **delivery pipe** with **delivery valve** ; **shaft, bearings, stuffing box / gland** .

1.4 Heads

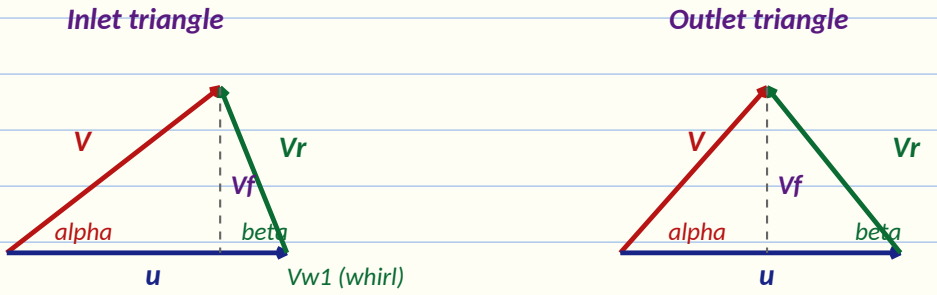
Term	Meaning
Suction head Hs	Vertical height of pump centre line above the sump water level
Delivery head Hd	Vertical height of the delivery outlet above the pump centre line
Static head Hst	Hs + Hd (total vertical lift)
Head loss	hfs (suction pipe) + hfd (delivery pipe) friction and minor losses
Manometric head Hm	Head actually developed by the pump: Hm = Hs + Hd + hfs + hfd + $Vd^2/2g$
Euler (theoretical) head He	Head given to the water by the impeller (before hydraulic losses)

$Hm = Hs + Hd + hfs + hfd + Vd^2 / 2g$; also $Hm = (\text{pressure head at delivery} - \text{pressure head at suction}) + (Vd^2 - Vs^2)/2g + \text{datum difference}$

2. Work Done by the Impeller

2.1 Velocity triangles

- **Inlet**: $u1 = \pi D1 N / 60$, Vf1 (radial), Vw1, Vr1, vane angle $\beta 1$ (with direction of u, backward). **Outlet**: $u2 = \pi D2 N / 60$, Vf2, Vw2, Vr2, outlet vane angle $\beta 2$.



Vw2 = u2 - Vf2 cot β2 (backward-curved vanes, β2 < 90°)

- For **radial entry** (no pre-whirl), α1 = 90°, Vw1 = 0. Flow rate **Q = π D1 B1 Vf1 = π D2 B2 Vf2** (B = width of impeller).

2.2 Euler's equation

Work done per second on water = ρ Q (Vw2 u2 - Vw1 u1)

Work done per unit weight (Euler head) He = (Vw2 u2 - Vw1 u1) / g ; for radial entry He = Vw2 u2 / g

- Backward vanes (β2 < 90°, 15° to 40°) give **stable operation and good efficiency**; radial vanes (β2 = 90°) give higher head but lower efficiency; forward vanes are not used.
- Pressure rise in the impeller (centrifugal head) = $(u2^2 - u1^2) / 2g + (Vr1^2 - Vr2^2) / 2g$.

2.3 Solved Problem 1

Impeller D2 = 0.4 m, N = 1450 rpm, B2 = 20 mm, β2 = 25°, Vf2 = 2.5 m/s, radial entry, ηman = 80%, ηo = 70%. Find Vw2, He, Hm, Q and shaft power.

- u2 = π x 0.4 x 1450 / 60 = **30.37 m/s** ; Vw2 = 30.37 - 2.5 x cot 25° = 30.37 - 5.36 = **25.0 m/s**.

He = Vw2 u2 / g = 25.0 x 30.37 / 9.81 = 77.4 m; Hm = 0.8 x 77.4 = *61.9 m*

Q = π D2 B2 Vf2 = π x 0.4 x 0.02 x 2.5 = 0.0628 m^3/s

Water power = ρ g Q Hm = 1000 x 9.81 x 0.0628 x 61.9 = 38.2 kW; Shaft power = 38.2 / 0.7 = *54.5 kW*

3. Losses and Efficiencies

3.1 Losses in a pump

Loss	Cause
Hydraulic losses	Friction in suction pipe, impeller and casing; shock (eddy) losses at inlet and outlet of vanes and at casing; diffusion losses
Mechanical losses	Friction in bearings, glands, stuffing box; disc friction (impeller rotating in fluid)

Leakage (volumetric) losses	Liquid flows back from delivery to suction through the gap between impeller and casing (wear ring clearance)

3.2 Efficiencies

Manometric efficiency $\eta_{man} = H_m / H_e = g\,H_m / (V_{w2}\,u_2)$

Volumetric efficiency $\eta_v = Q / (Q + q)$ (q = leakage) ; Mechanical efficiency η_m = (Power given to water by impeller) / (Shaft power)

Overall efficiency $\eta_o = (\rho\,g\,Q\,H_m)$ / (Shaft power) = $\eta_{man} \times \eta_v \times \eta_m$

- Typical overall efficiency of centrifugal pumps: 60 to 85% (large pumps up to 90%).
- Power input: $P = \rho\,g\,Q\,H_m / \eta_o$. Motor power is increased by 10 to 20% as a safety factor.

3.3 Solved Problem 2 (heads and power)

H_s = 4 m, H_d = 26 m, h_{fs} = 0.5 m, h_{fd} = 1.5 m, delivery velocity V_d = 2 m/s, Q = 0.05 m^3/s, η_o = 70%.

$H_m = 4 + 26 + 0.5 + 1.5 + 2^2 / 19.62 = 32.2$ m

$P = \rho\,g\,Q\,H_m / \eta_o = 1000 \times 9.81 \times 0.05 \times 32.2 / 0.70 = *22.6$ kW*

4. Minimum Starting Speed

4.1 Concept

- At start, the water in the impeller is **at rest and acts as a forced vortex**. The pump will start delivering only if the **pressure rise due to centrifugal action across the impeller $(u_2^2 - u_1^2)/2g$ is at least equal to the manometric head H_m**.

4.2 Derivation

Condition: $(u_2^2 - u_1^2) / 2g \geq H_m = \eta_{man}\,V_{w2}\,u_2 / g$ $\quad u_2^2 - u_1^2 = 2\,\eta_{man}\,V_{w2}\,u_2$

- Substituting $u = \pi\,D\,N / 60$: $(\pi\,N/60)^2\,(D_2^2 - D_1^2) = 2\,\eta_{man}\,V_{w2}\,(\pi\,D_2\,N / 60)$.

$N_{min} = 120\,\eta_{man}\,V_{w2}\,D_2 / [\,\pi\,(D_2^2 - D_1^2)\,]$ (rpm)

- Below N_{min} the pump cannot lift water to the delivery height (even if primed), so **the pump must run above this speed, with the delivery valve closed or partly open, at starting**.

Problem 3: D_2 = 0.4 m, D_1 = 0.2 m, V_{w2} = 15 m/s, η_{man} = 75%.

$N_{min} = 120 \times 0.75 \times 15 \times 0.4 / [\pi\,(0.16 - 0.04)] = 540 / 0.377 = *1432$ rpm*

5. Priming

- **Priming** = filling the suction pipe, casing and impeller **completely with the liquid** (expelling the air) before starting. Because air has very low density, a pump full of air produces negligible pressure rise and cannot lift water (**air binding**).
- **Methods**: pouring water through a funnel at the top of the casing (with foot valve), **self-priming pumps** , **ejector** , **vacuum pump** , jockey pump, or placing the pump **below** the water level (flooded suction, no priming needed).
- Starting sequence: prime, close the delivery valve (minimum power), start motor, open delivery valve slowly. At shut-off, close the delivery valve first.

6. Specific Speed

$N_s = N\sqrt{Q} / Hm^{(3/4)}$ (N in rpm, Q in m^3/s, Hm in m)

- **Specific speed of a pump = the speed of a geometrically similar pump that delivers 1 m^3/s against a head of 1 m.** It characterises the **shape** of the impeller.

Ns (approx., SI)	Pump type
10 to 30	Low-speed radial flow (narrow impeller, high head, low discharge)
30 to 80	Medium-speed radial (Francis-type)
80 to 160	High-speed / mixed flow
> 160	Axial flow (propeller) pumps: low head, large discharge

- Double-suction pump: use Q/2 in Ns. For multistage pumps, use the head per stage.

Problem 4: N = 1450 rpm, Q = 0.0628 m^3/s, Hm = 61.9 m. $N_s = 1450 \times \sqrt{0.0628} / 61.9^{0.75} = 1450 \times 0.2506 / 22.07 =$ **16.5** (low-speed radial impeller).

7. Limitation of Suction Lift and NPSH

7.1 Maximum suction lift

- Theoretically the suction lift cannot exceed **atmospheric head Ha = 10.33 m of water**. In practice it is limited to **about 6 to 7 m** because of vapour pressure Hv, friction losses hfs and velocity head in the suction pipe, and the pressure drop at the impeller entrance (cavitation).

Hs(max) = Ha - Hv - hfs - NPSHR (pump above the sump level)

7.2 Net Positive Suction Head

NPSH = total absolute head at the pump suction flange above the vapour pressure head, referred to the pump axis.

NPSH available (NPSHA) = Ha - Hv - Hs - hfs (Hs = suction lift; use - Hs for a lift and + Hs for flooded suction)

NPSH required (NPSHR) = a property of the pump (from the manufacturer's test curves; increases with Q and N)

- **Condition for cavitation-free operation: NPSHA > NPSHR** (with a margin of 0.5 to 1 m).

Thoma's cavitation number σ = NPSH / Hm (compare with σc)

- Remedies if NPSHA is low: lower the pump (reduce Hs), **increase pipe diameter** (reduce hfs), reduce flow temperature, reduce speed, use **inducer** or a pump with lower NPSHR, increase sump level.

Problem 5: Ha = 10.3 m, Hv = 0.24 m (water at 20°C), Hs = 3.5 m, hfs = 0.8 m, NPSHR = 4 m.

NPSHA = 10.3 - 0.24 - 3.5 - 0.8 = 5.76 m (> 4 m, safe margin 1.76 m)

Maximum suction lift = 10.3 - 0.24 - 0.8 - 4 = 5.26 m

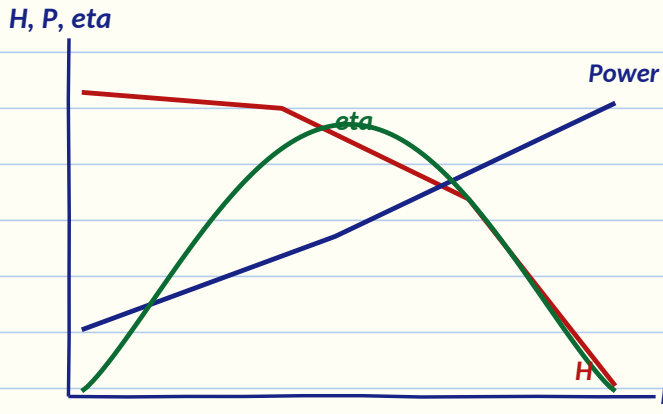
8. Cavitation in Pumps

- **Cavitation**: vapour bubbles form in the low-pressure region (impeller eye, back of vanes) when pressure falls to vapour pressure; they collapse as they move to high pressure.
- **Effects**: **noise and vibration** (crackling, gravel-like sound); **pitting and erosion** of impeller and casing; **fall in head, discharge and efficiency**; reduced pump life; possible shaft and bearing damage.
- **Prevention**: maintain NPSHA > NPSHR; low suction lift and short, straight, large suction pipe; avoid air leaks and sharp bends; use cavitation-resistant impeller materials (bronze, stainless steel); operate near the best efficiency point; keep liquid temperature low.

9. Performance and Characteristic Curves

9.1 Types of curves

Curve	Plotted	Constant
Main characteristic curves	Hm, P, η against Q at several speeds	Speed N varied
Operating characteristic curves	Hm, P, η against Q	Speed N constant
Constant efficiency (Muschel) curves	Contours of η on Hm-Q plane	η



Main characteristic curves:
H-Q falls as Q rises.
Power-Q rises with Q.
Efficiency peaks at the **best efficiency point (BEP).**

- **Hm-Q** : falls as Q rises (for backward vanes). **P-Q** : rises with Q (so start with delivery valve closed). **η-Q** : rises to a maximum (**best efficiency point, BEP**) then falls.
- The **shut-off head** is at Q = 0; the **runout** is at maximum flow.

9.2 Affinity laws (similarity)

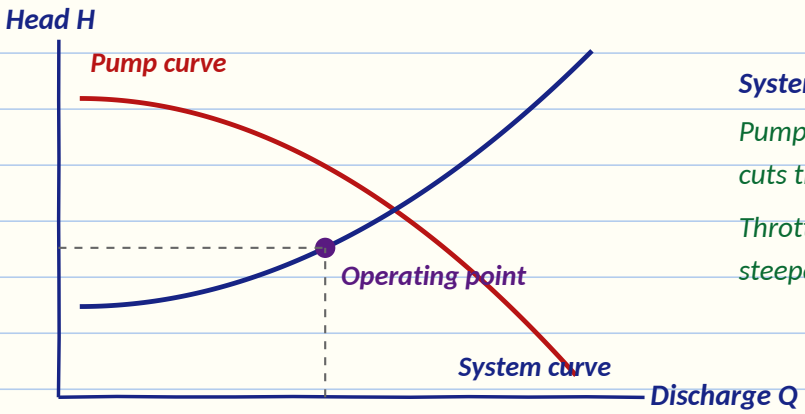
$Q \propto N D^3$; $Hm \propto N^2 D^2$; $P \propto N^3 D^5$ (same pump: $Q \propto N$, $H \propto N^2$, $P \propto N^3$)

Unit quantities: Nu = N / √Hm ; Qu = Q / √Hm ; Pu = P / Hm^(3/2)

Problem 6: At 1450 rpm a pump gives Q = 0.06 m^3/s, H = 40 m, P = 30 kW. Find values at 1750 rpm.

- Ratio r = 1750 / 1450 = 1.207. Q = 0.06 x 1.207 = **0.0724 m^3/s** ; H = 40 x 1.457 = **58.3 m** ; P = 30 x 1.758 = **52.7 kW** .

9.3 Operating point (pump and system)



System head = static head + K Q^2

Pump works where its H-Q curve cuts the system curve.

Throttling the delivery valve steepens the system curve.

- **System curve**: $H_{sys} = Hst + K Q^2$ (static head plus friction). The **operating point** is the intersection of pump and system curves.

Problem 7: Pump curve $H = 50 - 1500 Q^2$; system $H = 20 + 2000 Q^2$. Equate: $3500 Q^2 = 30$, so $Q^2 = 0.00857$, **$Q = 0.0926\ m^3/s$** *and* **$H = 37.1\ m$**.

10. Multistage Centrifugal Pumps

10.1 Need

- A single impeller gives limited head (about 80 to 100 m practical). For **high heads** (boiler feed, mines, high-rise buildings), several impellers are used in a single casing.

10.2 Pumps in series (multistage for head)



- Impellers are mounted on the **same shaft**, the **discharge of one stage enters the next**; **same Q, head adds**: $H = n \times H_{stage}$. **Guide vanes (diffusers)** between stages convert velocity to pressure and lead water to the next impeller eye.

10.3 Pumps in parallel (for discharge)

- Two or more impellers (or pumps) deliver into a **common delivery pipe**: **same head, discharge adds**: $Q = Q1 + Q2 + \ldots$ Used for large discharge at moderate head (low lift pumps); double-suction impeller is a parallel arrangement.

Arrangement	Head	Discharge	Use
Series (multistage)	Adds (n x H)	Same	High head

Parallel	Same	Adds (n x Q)	High discharge, standby

Problem 8: H = 120 m, Q = 0.05 m^3/s, N = 1450 rpm. Single stage: Ns = 1450 x √0.05 / 120^0.75 = 1450 x 0.2236 / 36.26 = 8.9 (impractically low). With 4 stages (30 m each): Ns = 1450 x 0.2236 / 30^0.75 = 1450 x 0.2236 / 12.82 = **25.3**, a good efficient radial impeller.

11. Troubles and Remedies

Trouble	Probable cause	Remedy
Pump does not deliver water	Not primed, air leak in suction, foot valve stuck, speed too low, suction lift too high	Re-prime, seal leaks, repair foot valve, check speed and rotation, reduce lift
Low discharge / head	Wrong direction of rotation, worn impeller or wear rings, partly clogged strainer, air entering gland	Correct rotation, replace impeller / rings, clean strainer, repack gland
Loses water after starting	Air leak, low NPSH, water level in sump falls	Fix leaks, raise sump level, increase pipe size
Motor overload / excess power	Speed too high, delivery valve fully open at low head, mechanical binding	Throttle delivery valve, check speed, realign shaft
Vibration and noise	Cavitation, misalignment, unbalanced impeller, loose foundation, damaged bearing	Raise NPSHA, align shaft, balance impeller, tighten bolts, replace bearing
Overheating of bearings / pump	Poor lubrication, running with valve closed, misalignment	Lubricate, avoid shut-off running, align
Leakage at gland	Worn packing or sleeve	Replace packing, tighten gland evenly
Pitting of impeller	Cavitation, corrosion	Improve suction conditions, use better material

12. More Concepts and Solved Problems

12.1 Centrifugal vs reciprocating pump

Basis	Centrifugal pump	Reciprocating pump
Discharge	Large, continuous and steady	Small, pulsating (needs air vessel)
Head	Low to medium (multistage for high)	Very high heads possible
Viscous / dirty liquid	Suitable for clean, moderately dirty	Not suitable for dirty liquids
Efficiency curve	Falls when head rises	Nearly constant (positive displacement)
Cost, space, maintenance	Compact, low cost, simple	Bulky, costly, more wear parts
Priming	Required (except self-priming)	Self-priming

12.2 Casing types

Casing	Features
Volute	Spiral passage with area increasing towards the delivery; velocity head converted to pressure; simple, common
Vortex	Circular chamber (vortex) between impeller and volute; improves efficiency slightly; larger size
Diffuser (turbine pump)	Fixed guide vanes surround the impeller; smoother conversion; used in multistage and high-head pumps

12.3 Mixed flow and axial flow pumps

- **Mixed flow pump**: liquid leaves the impeller partly radially and partly axially (Ns 80 to 160): medium head and large discharge. **Axial flow (propeller) pump**: flow parallel to the shaft (Ns > 160): large discharge, low head (drainage, irrigation, cooling water); has a rising power curve at low flow and a steep head curve.

12.4 Slip and slip factor

- The actual whirl velocity at outlet is **less than the theoretical** V_{w2} because of relative circulation of liquid in the impeller passages (**slip**). Slip factor $\sigma_s = V_{w2}' / V_{w2} < 1$; **Stodola**: $\sigma_s = 1 - 0.63\pi / Z$ (Z = number of vanes). For $Z = 8$: $\sigma_s = 0.75$.

12.5 Effect of vane outlet angle

Vane type	β_2	Head	Remarks

Backward curved	< 90°	Lower Euler head	Stable, high efficiency, power curve self-limiting
Radial	= 90°	Moderate	Used in small pumps and slurry service
Forward curved	> 90°	Highest Euler head	Poor efficiency, unstable, large kinetic energy loss

Problem 9: Impeller design

N = 1450 rpm, Hm = 40 m, Q = 0.1 m^3/s, ηman = 80%, speed coefficient $u2 = \sqrt{(2\,g\,Hm)}$, $Vf2 = 0.12\sqrt{(2\,g\,Hm)}$. Find D2, Vw2, β2 and B2.

- $\sqrt{(2\,g\,Hm)} = 28.0$ m/s, so u2 = 28.0 m/s; $D2 = 60\,u2 / \pi N = 60 \times 28.0 / (\pi \times 1450) =$ **0.369 m**.
- $He = Hm / \eta man = 50$ m; $Vw2 = g\,He / u2 = 9.81 \times 50 / 28.0 =$ **17.5 m/s**; Vf2 = 3.36 m/s.

$\tan \beta 2 = Vf2 / (u2 - Vw2) = 3.36 / 10.5 = 0.32 \quad \beta 2 = 17.7°$

$B2 = Q / (\pi\, D2\, Vf2) = 0.1 / (\pi \times 0.369 \times 3.36) = 0.0257$ m ≈ 26 mm

Problem 10: Pumps in parallel and series

Each pump: $H = 50 - 1500\,Q^2$; system: $H = 20 + 2000\,Q^2$. Find the operating point for (a) one pump, (b) two in parallel, (c) two in series.

- (a) Single: Q = 0.0926 m^3/s, H = 37.1 m.

(b) Parallel: $H = 50 - 1500\,(Q/2)^2 = 50 - 375\,Q^2$; equate with system: $30 = 2375\,Q^2$, Q = 0.112 m^3/s, H = 45.3 m

(c) Series: $H = 100 - 3000\,Q^2$; $80 = 5000\,Q^2$, Q = 0.1265 m^3/s, H = 52 m

- **Note**: discharge is not doubled in parallel because friction head increases with Q^2; series raises head, but flow rises only moderately.

Problem 11: Power and energy

Lift 100 m^3/h to a tank, Hm = 25 m, ηo = 65%.

- Q = 100/3600 = 0.02778 m^3/s; $P = 1000 \times 9.81 \times 0.02778 \times 25 / 0.65 =$ **10.5 kW**; energy for 8 h = 84 kWh.

Problem 12: Specific speed of double-suction pump

N = 980 rpm, Q = 0.4 m^3/s, Hm = 25 m, double suction (each side Q/2 = 0.2 m^3/s).

Ns = 980 x √0.2 / 25^0.75 = 980 x 0.447 / 11.18 = *39.2* (medium-speed radial pump)

Problem 13: Centrifugal pressure rise

Data of Problem 1 with D1 = 0.2 m: u1 = 15.2 m/s, u2 = 30.4 m/s.

Centrifugal head = $(u2^2 - u1^2) / 2g = (922 - 231) / 19.62 = 35.2$ m (compare with Hm; pump starts only if this is more than Hm, otherwise needs a higher speed, as in Problem 3).

13. Quick Revision and Important Questions

Euler: He = Vw2u2/g; Vw2 = u2 - Vf2 cotβ2. Hm = Hs + Hd + losses + $Vd^2/2g$. ηo = ηman ηv ηm. Nmin = 120ηman Vw2 D2/[π($D2^2$ - $D1^2$)]. Ns = N√Q/Hm^(3/4). NPSHA = Ha - Hv - Hs - hfs > NPSHR. Affinity: Q N, H N^2, P N^3.

- Explain the working of a centrifugal pump with a neat sketch. State its main parts. (10 M)
- Derive the expression for the work done by the impeller (Euler's head equation) with velocity triangles. (10 M)
- Define manometric head, and manometric, volumetric, mechanical and overall efficiencies. (5 M)
- Derive the expression for the minimum starting speed of a centrifugal pump. (5 M)
- What is priming? What is NPSH? Explain the limitation of suction lift. (5 M)
- Define specific speed. Explain performance (characteristic) curves of a centrifugal pump. (10 M)
- Write a note on cavitation in pumps and multistage pumps. List common troubles and remedies. (10 M)